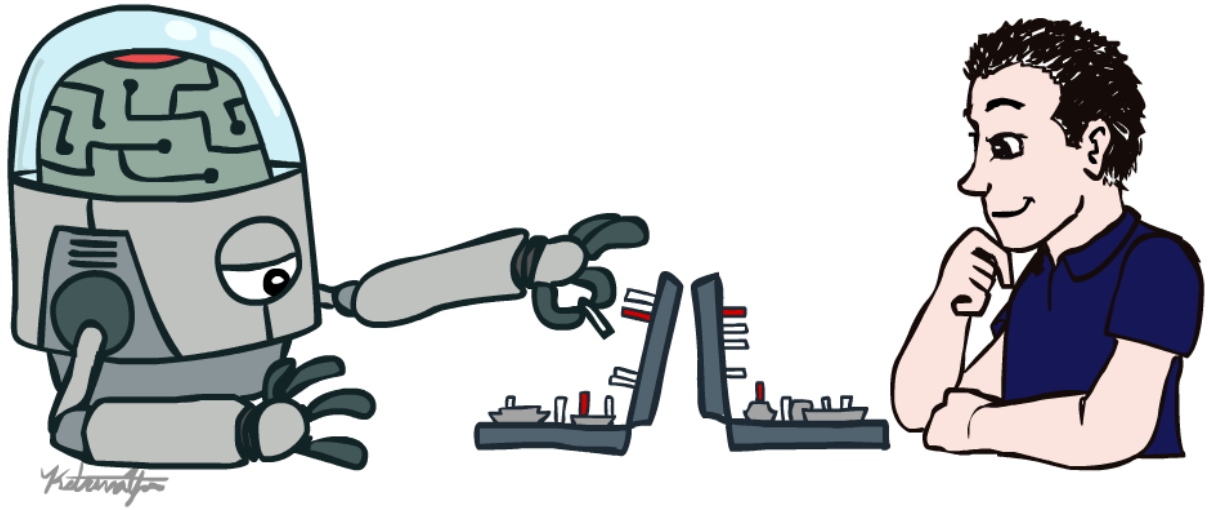
CS 188: Artificial Intelligence

Introduction



Fall 2026

University of California, Berkeley

Peyrin Kao and Emma Pierson

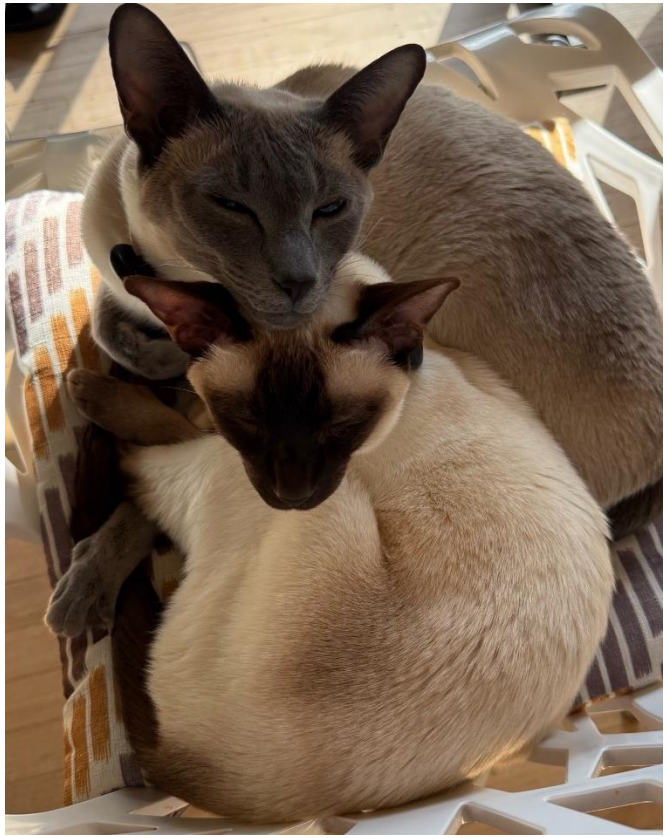
[These slides are based on those created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley (ai.berkeley.edu).]

Beginning of Today: Intro and Logistics

- Staff introductions: Peyrin, Emma, and course staff
- Course logistics
  - Discuss briefly, full details on course website

Staff Introductions: Emma (she/her)

- Did my undergrad, master's, and PhD at Stanford (sorry)
  - Also spent a year at Oxford (happy to chat about studying overseas)
- Assistant Professor of Computer Science (either Emma or Professor Pierson are fine)
  - My lab develops AI methods for the health and social sciences, focusing on improving health and reducing inequality
  - Examples: using AI to map migration in response to wildfires; detect flooding; understand inequality in pain; quantify gender, racial, and other social biases; forecast the spread of COVID-19
  - Affiliations: Berkeley AI Research Lab (BAIR), Computational Precision Health
- Other loves besides teaching/research
  - Writing for a mass audience; being outside; classical piano; my cats Sigma and Mew



Staff Introductions: Peyrin (he/him)

- TA (2018-2022), lecturer (2022-present)
- https://peyrin.github.io



Actual real picture of me.

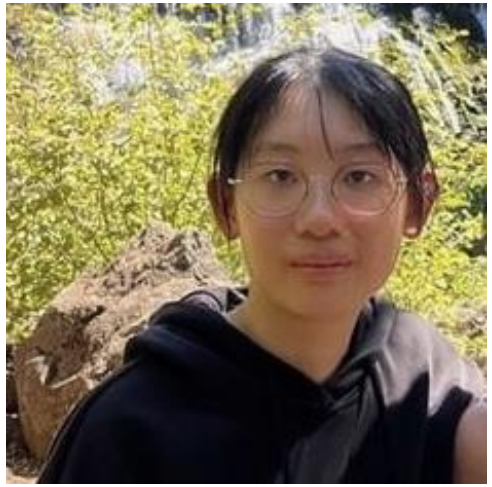
Our talented course staff!



Saathvik Selvan



Lima Ondine



Yinqi Yi



Surya Krishnapillai



Krish Patel



Michael Li

Enrollment

- We will likely be able to clear the waitlist and approve all concurrent enrollment requests within the next week, but no promises until it happens.
  - We have no further updates than this, so please don't ask.
- https://inst.eecs.berkeley.edu/~cs188/archive/sp26/fa26-faqs/ FAQ page if you have more questions

Course Structure: Lectures

- You are here!
- Mon/Wed, 2:00-3:30 PM PT
- Attendance is not taken
  - But there may be a bit of extra credit for attending some lectures
- You can attend:
  - In-person in Gateway 1210
  - Remotely over Zoom (we'll try our best to livestream)
  - Asynchronously by watching recordings (posted on website)

Course Structure: Exams

- Save the dates!
  - Midterm: Mon Oct 19, 7–9 PM PT
  - Final exam: Thu Dec 17, 3–6 PM PT
- If you can’t make it:
  - No remote exams
  - We’ll offer an in-person-only alternate exam right after the listed time
- More logistics closer to the exam

Resources

- Course website: https://inst.eecs.berkeley.edu/~cs188/fa26/
- All resources (slides, notes, recordings, assignments, etc.) posted here
- Ed: Discussion forum
- Staff email for private concerns: cs188@berkeley.edu
  - Making a private post on Ed is easier/faster
- Gradescope: Submit assignments here

Grading Structure

- Please see course website

Extensions and Accommodations

- We'll drop your lowest homework score
- If you ever need an extension, please request one!
  - We're here to support you, and we understand that life happens.
  - Extension form will be posted on the website

DSP

- Disabled Students' Program (DSP)
  - There's a variety of accommodations UC Berkeley can help us set up for you in this class
  - https://dsp.berkeley.edu/
- Please submit your DSP letter as soon as possible (if you have one)
- Our goal is to teach you the material in our course. The more accessible we can make it, the better.

Collaboration and Academic Dishonesty

- We're here to help! There are plenty of staff and resources available for you
  - You can always talk to a staff member if you're feeling stressed or tempted to cheat
  - Collaboration on homework is okay, but please cite collaborators
  - Do not post solutions online or share with others!
- Academic dishonesty policies
  - Reported to Center of Student Conduct
  - Negative points on assignments, and/or F in the class

GenAI Policy

- You are allowed to use GenAI on projects
- It is essential that you **understand everything you submitted**, and we will test this on exams (and potentially with other evaluations throughout the course)
- Please read GenAI policy on website for full details

"Our results show that having GPT-4 access while solving problems significantly improves performance…However, we additionally find that when access is subsequently taken away, students actually perform worse than those who never had access…**unfettered access to GPT-4 can harm educational outcomes.**"

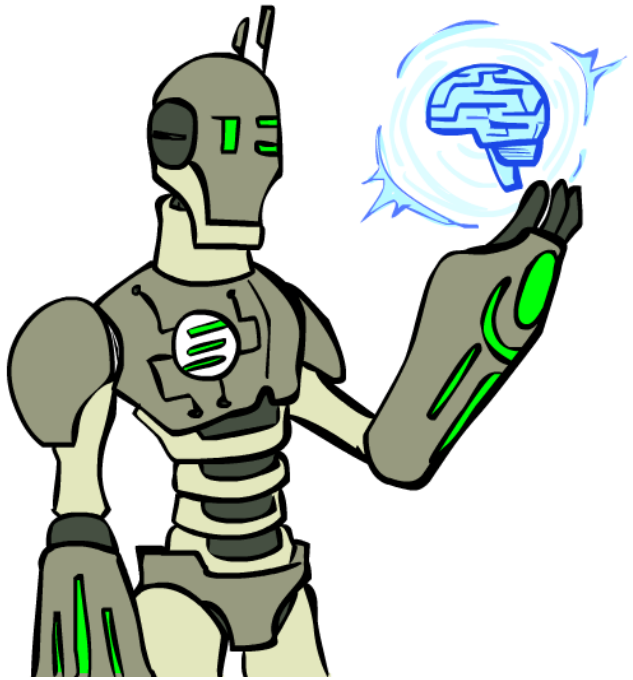
Bastani et al. *Proceedings of the National Academy of Sciences*, 2025.

Stress Management and Mental Health

- **Your health is more important than this course**
- If you feel overwhelmed, there are options
  - Academically: Ask on Ed, talk to staff in office hours, set up a meeting with staff to make a plan for your success this semester
  - Non-academic:
    - Counselling and Psychological Services (CAPS) has multiple free, confidential services
      - Casual consultations: https://uhs.berkeley.edu/counseling/lets-talk
      - Crisis management: https://uhs.berkeley.edu/counseling/urgent
    - Check out UHS's resources: https://uhs.berkeley.edu/health-topics/mental-health

Second Half of Today: What is AI?

- What is AI?
- What is this course?



AI is having real-world impact

- Public imagination
  - Text assistants



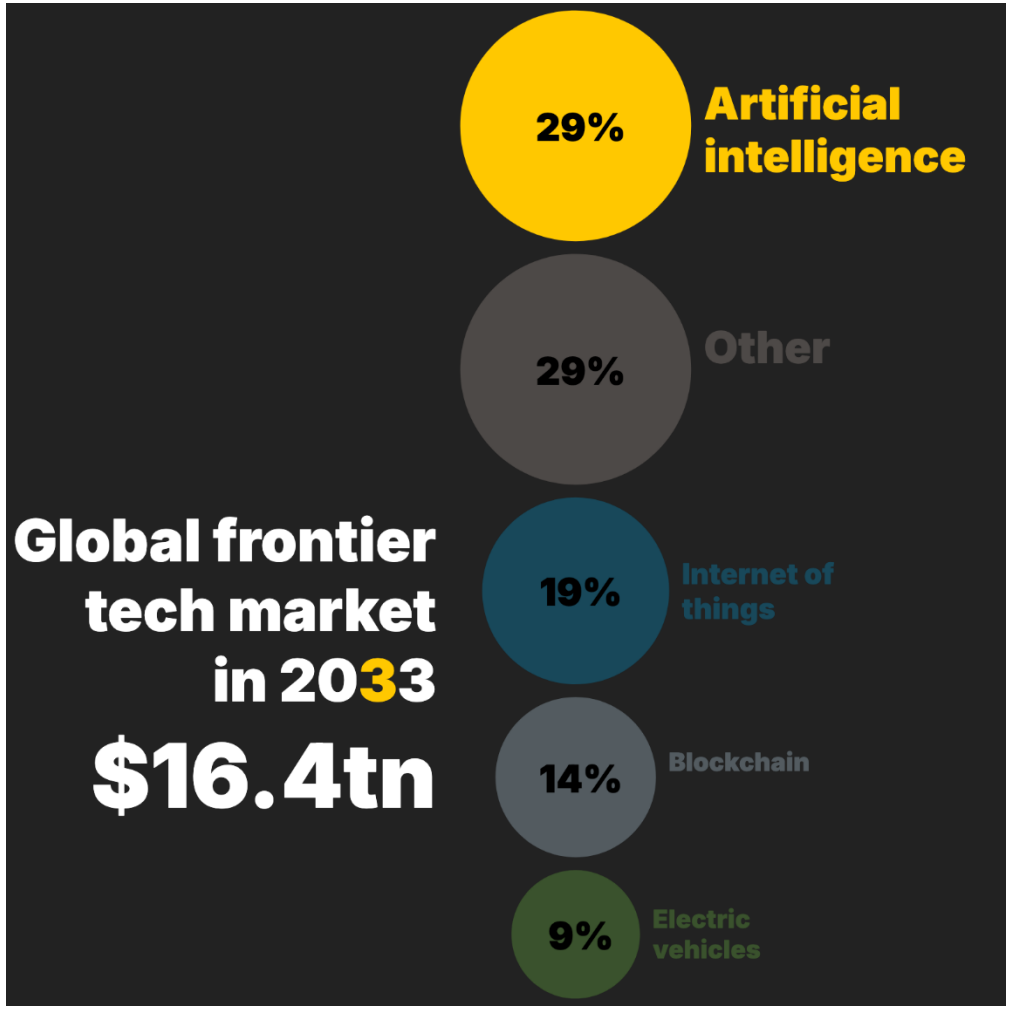
AI is having real-world impact

- Public imagination
  - Text assistants
  - Image generation



AI is having real-world impact

- Public imagination
- Economy
  - $4.8 trillion by 2033
    (25x increase in a decade)



https://unctad.org/publication/technology-and-innovation-report-2025

AI is having real-world impact

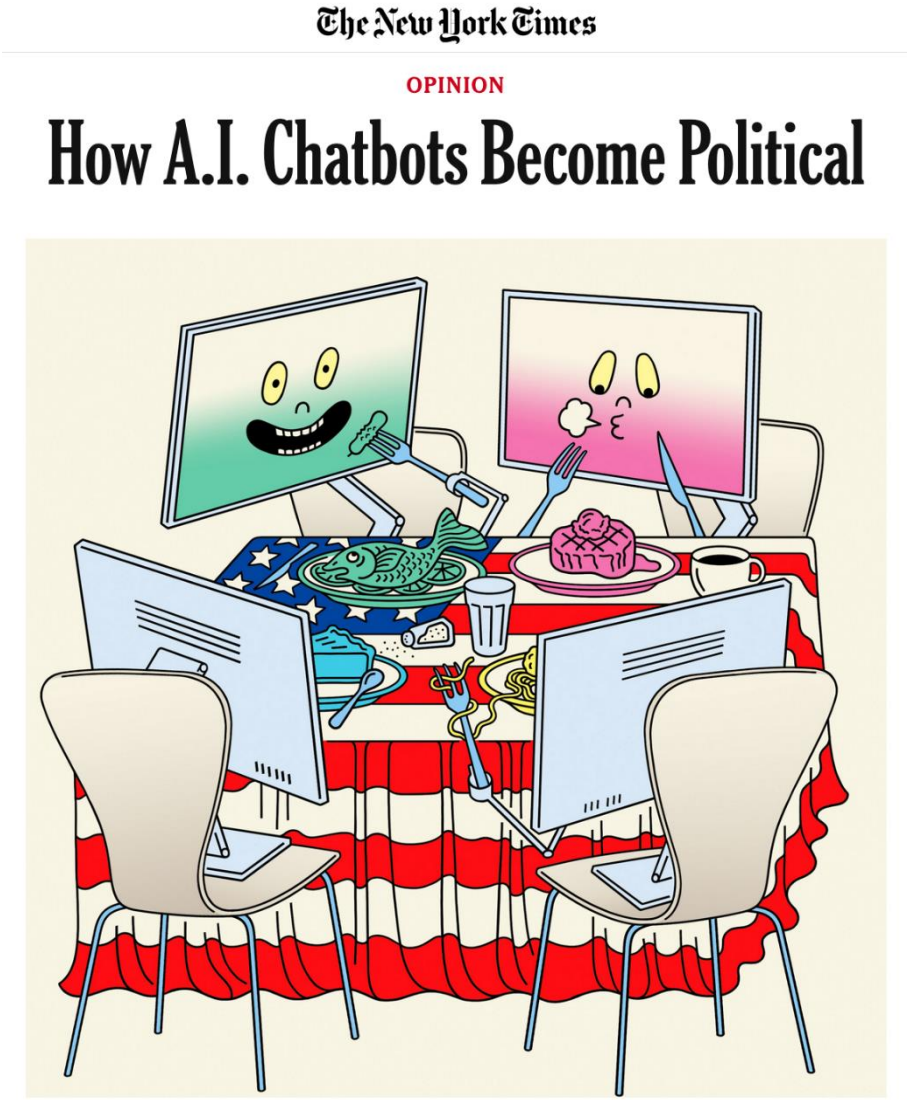
- Public imagination
- Economy
- Politics



https://www.nytimes.com/2025/02/11/world/europe/vance-speech-paris-ai-summit.html

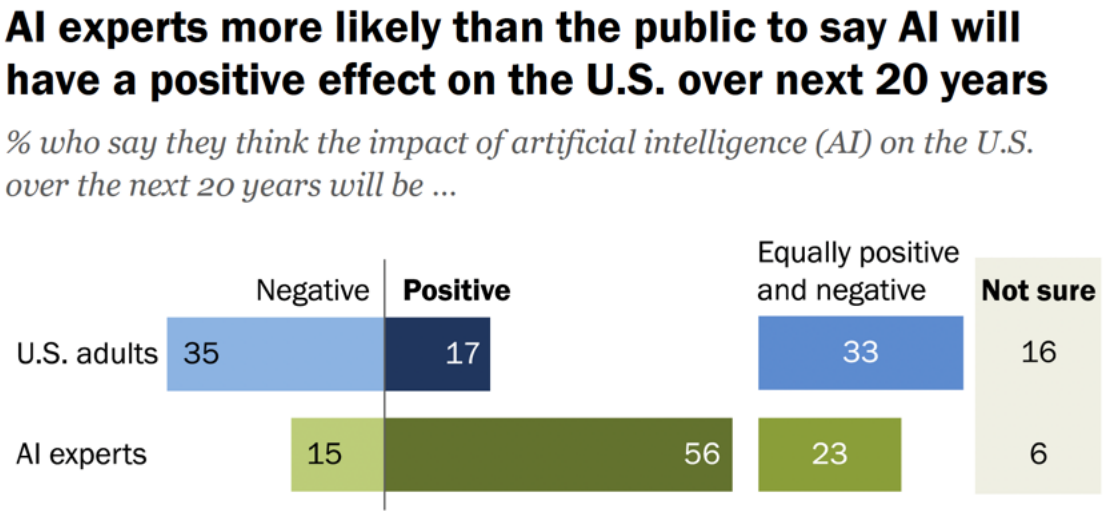
AI is having real-world impact

- Public imagination
- Economy
- Politics



AI is having real-world impact

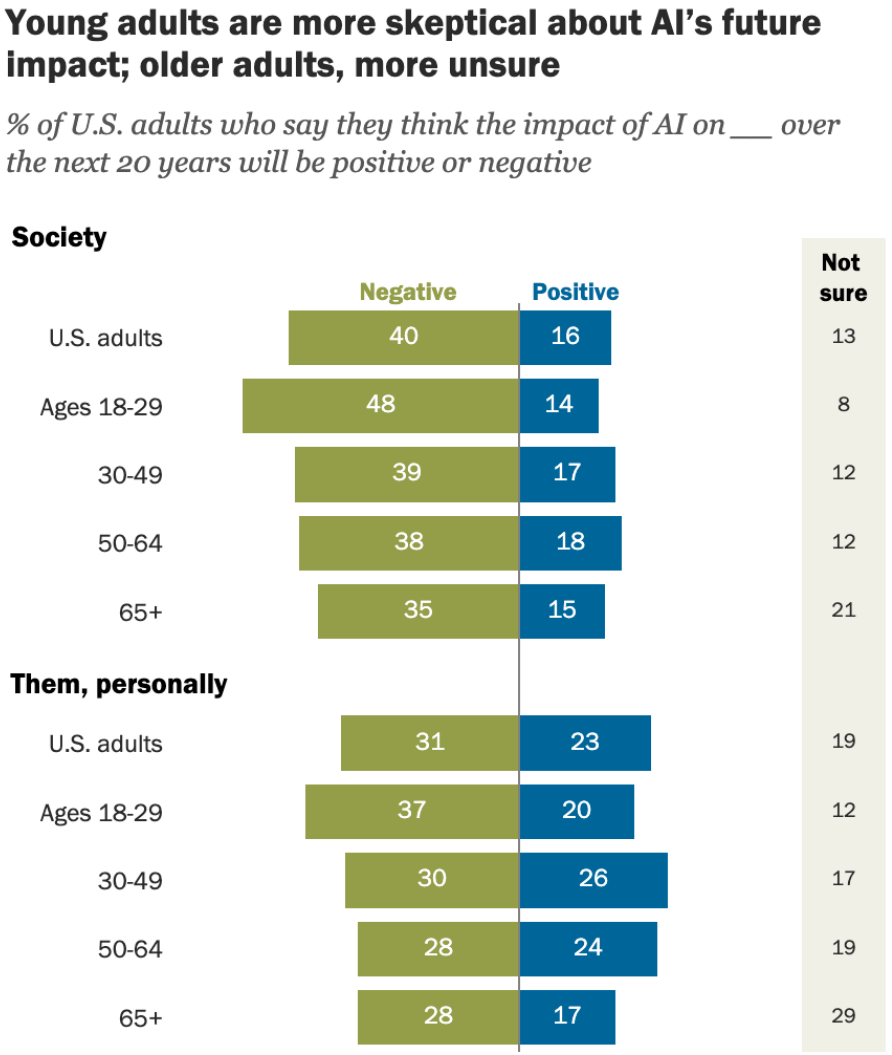
- Public imagination
- Economy
- Politics



Pew Research, 2025

AI is having real-world impact

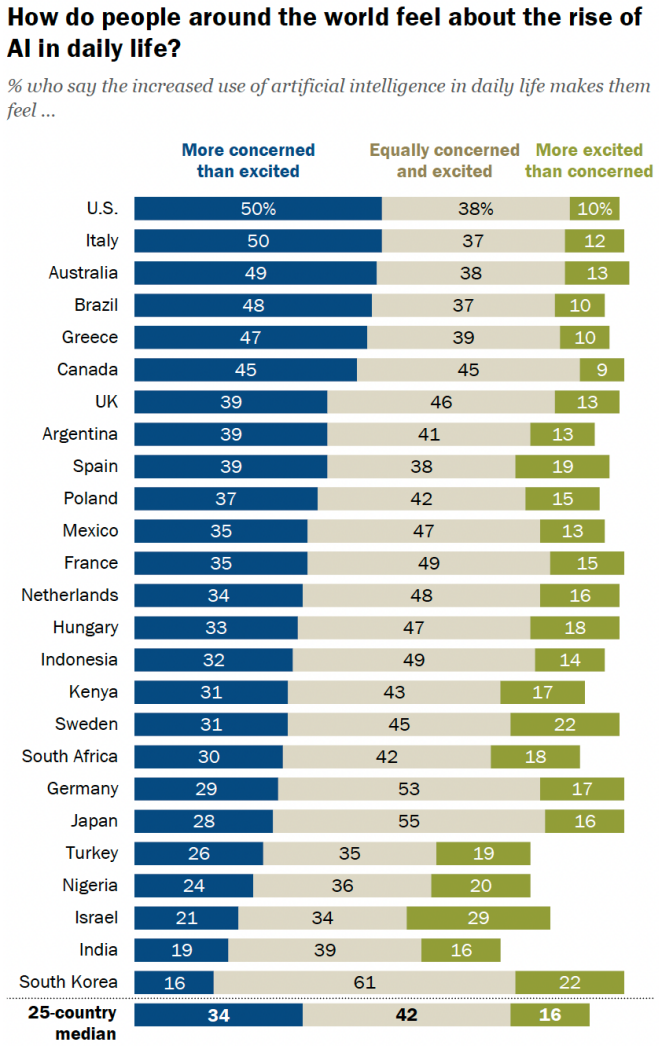
- Public imagination
- Economy
- Politics



Pew Research, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics



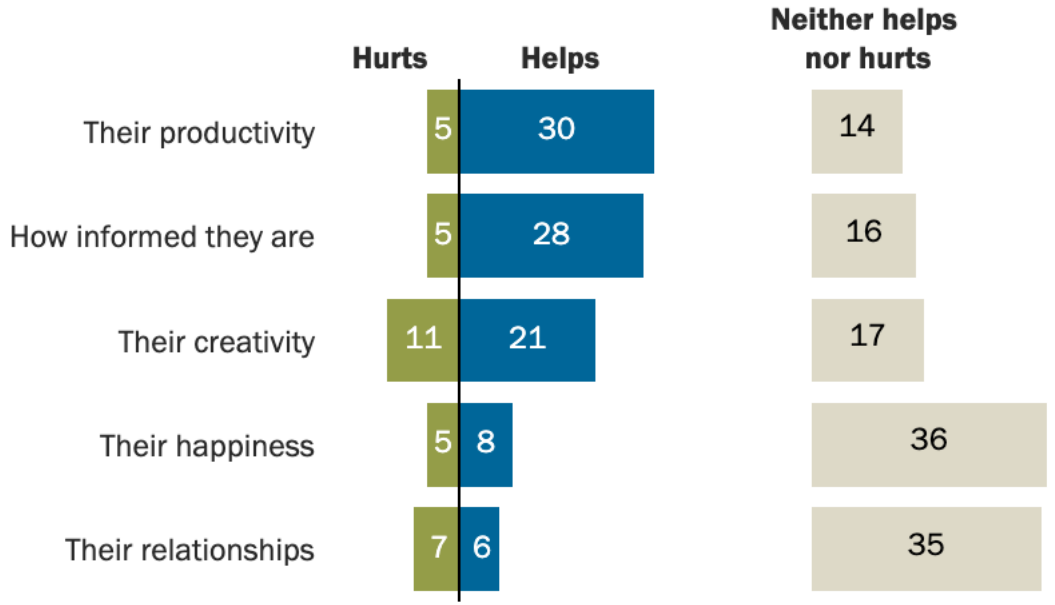
Pew Research, 2025

AI is having real-world impact

- Public imagination
- Economy
- Politics

Americans more likely to say chatbots help rather than hurt their productivity, knowledge and creativity

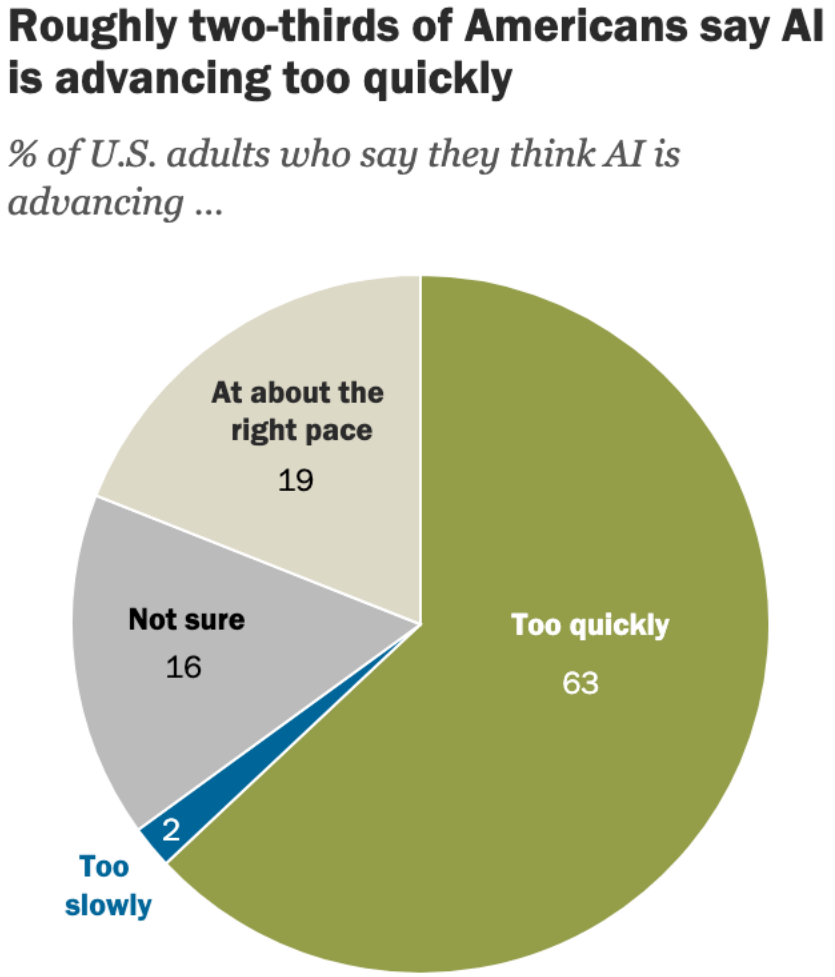
% of U.S. adults who say they think using AI chatbots ___ each of the following



Pew Research, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics



Pew Research, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics

🌻 no data centers in my backyard

money, power, and populism in the AI buildout

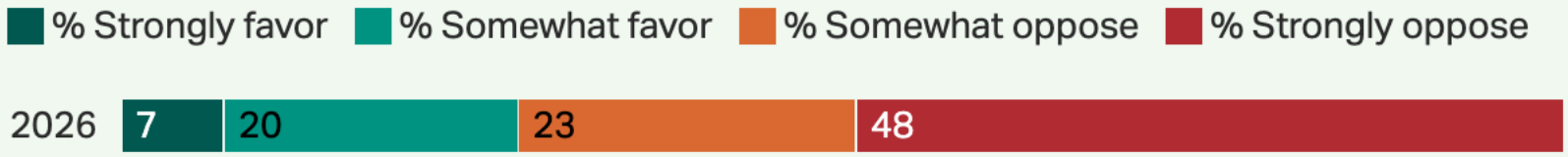


JASMINE SUN

AUG 03, 2026

Seven in 10 Americans Oppose Local Construction of AI Data Centers

Overall, would you strongly favor, somewhat favor, somewhat oppose or strongly oppose the construction of a data center in your area to support artificial intelligence, or AI, technology in the U.S.?



March 2-18, 2026
"No opinion" percentage not shown

Gallup, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics

Bernie Sanders warns ‘worst is yet to come’ in rallying cry against billionaires

US senator appears at Manhattan rally alongside New York City mayor Zohran Mamdani, who cautioned that AI is ‘coming for human jobs’

The Guardian, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law

Aug. 18, 2023, 12:18 PM; Updated: Aug. 18, 2023, 12:48 PM

AI-Generated Art Lacks Copyright Protection, D.C. Court Says (1)



Riddhi Setty
Reporter

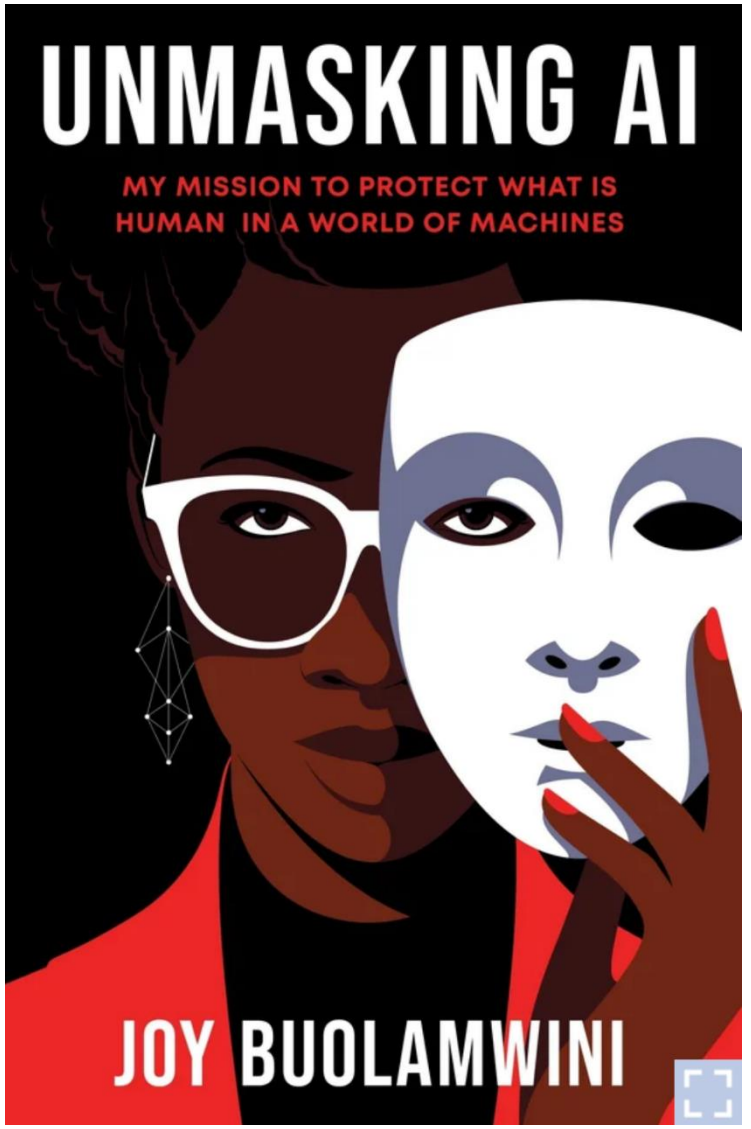


Isaiah Poritz
Legal Reporter

Bloomberg Law, 2023

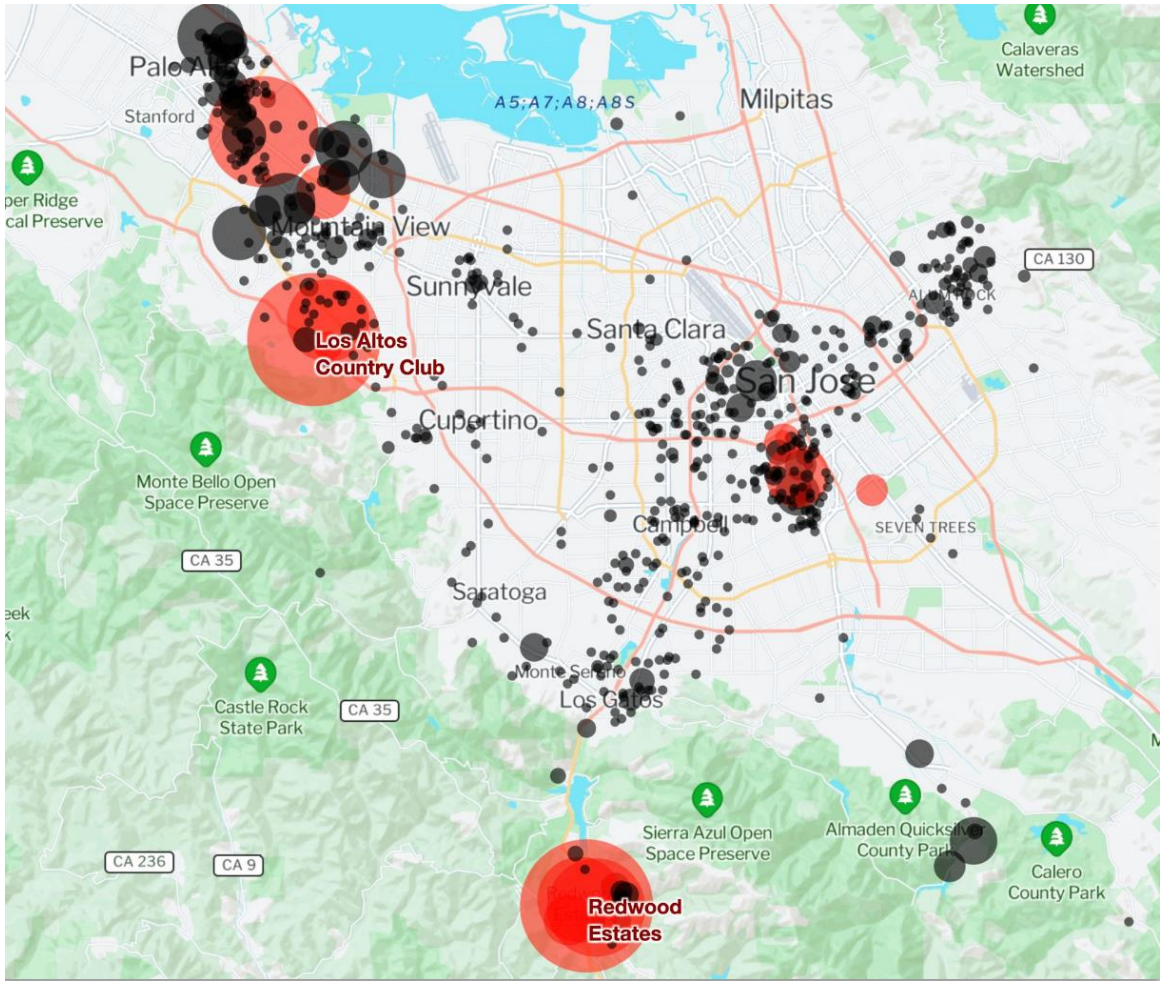
AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law



AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law



AI for Scaling Legal Reform: Mapping and Redacting Racial Covenants in Santa Clara County. Surani et al, 2024.

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law



https://dataplusfeminism.mit.edu

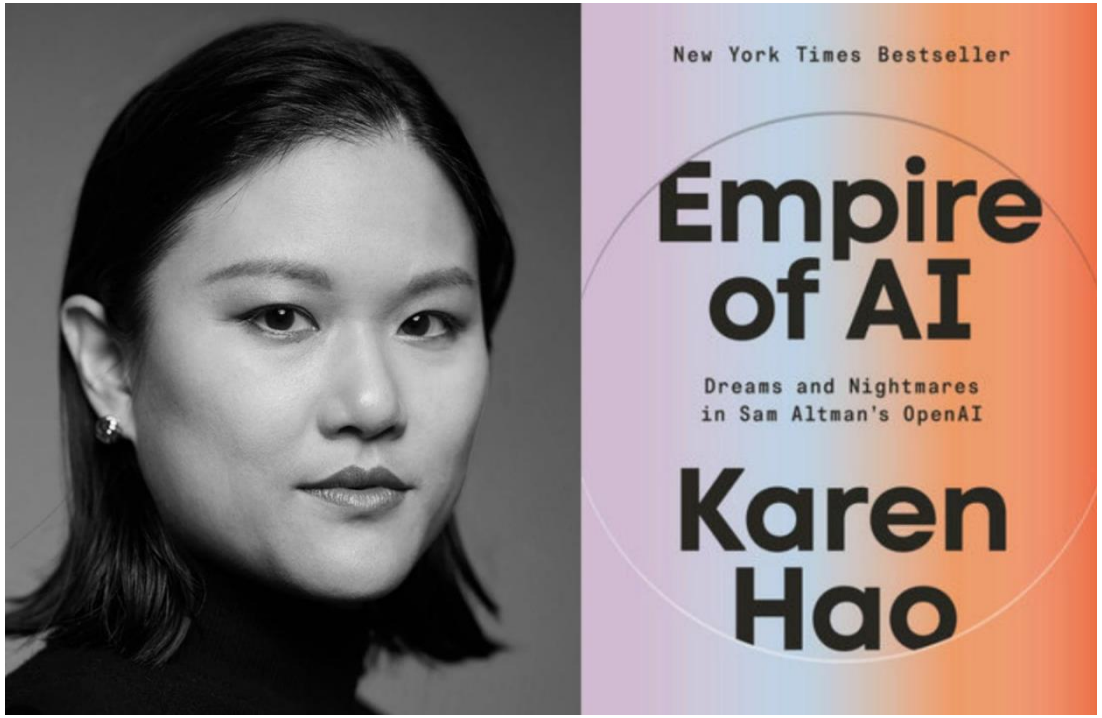
AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor

The human labor behind AI chatbots and other smart tools

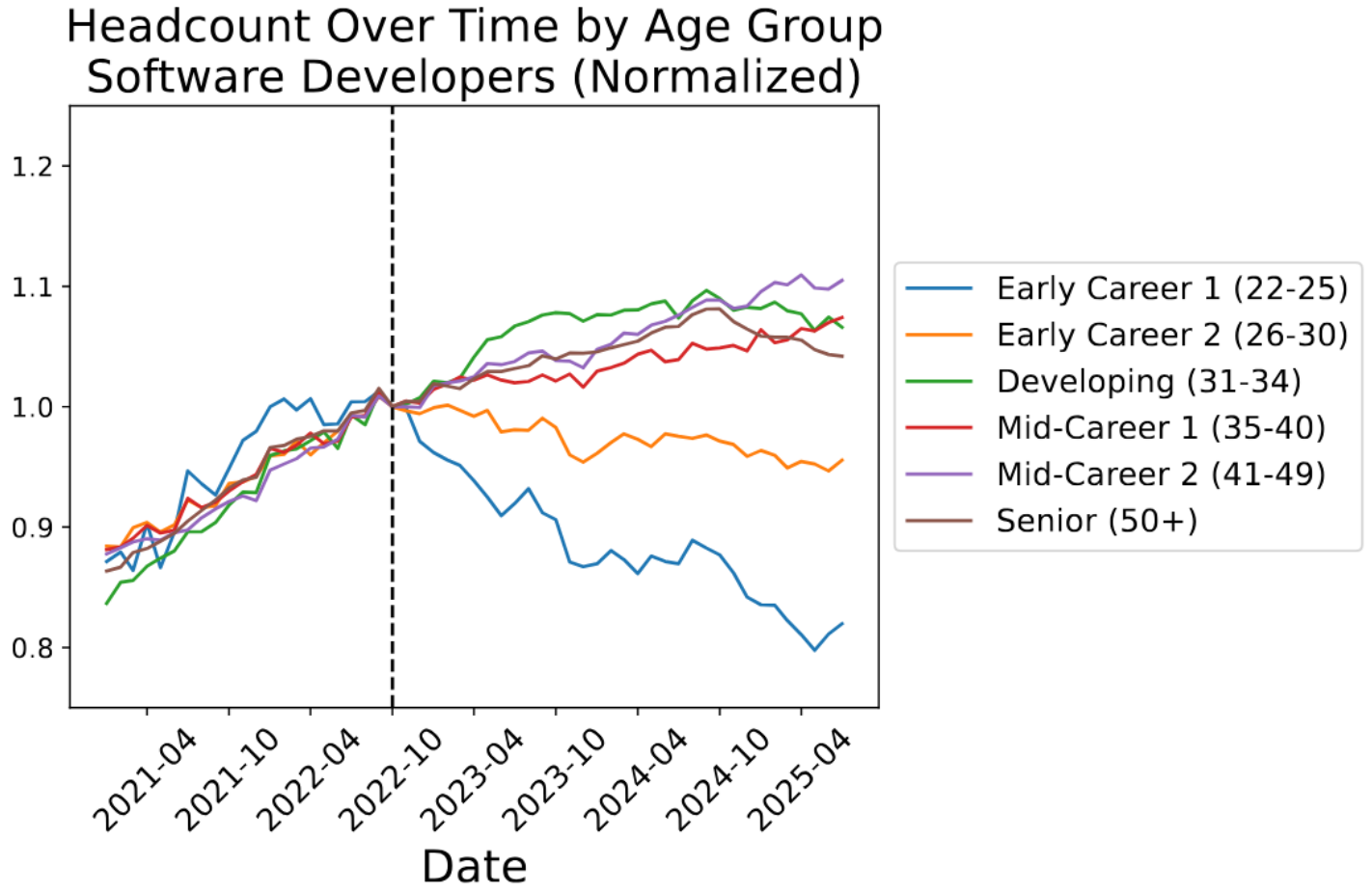
Data labeling is an important step in developing artificial intelligence but also exposes the people doing the work to harmful content.

MarketWatch, 2023



AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor



Stanford Digital Economy Lab, 2025

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Chemistry and biology

Science & technology | The 2024 Nobel prizes

AI wins big at the Nobels

Awards went to the discoverers of micro-RNA, pioneers of artificial-intelligence models and those using them for protein-structure prediction

The Economist, 2024

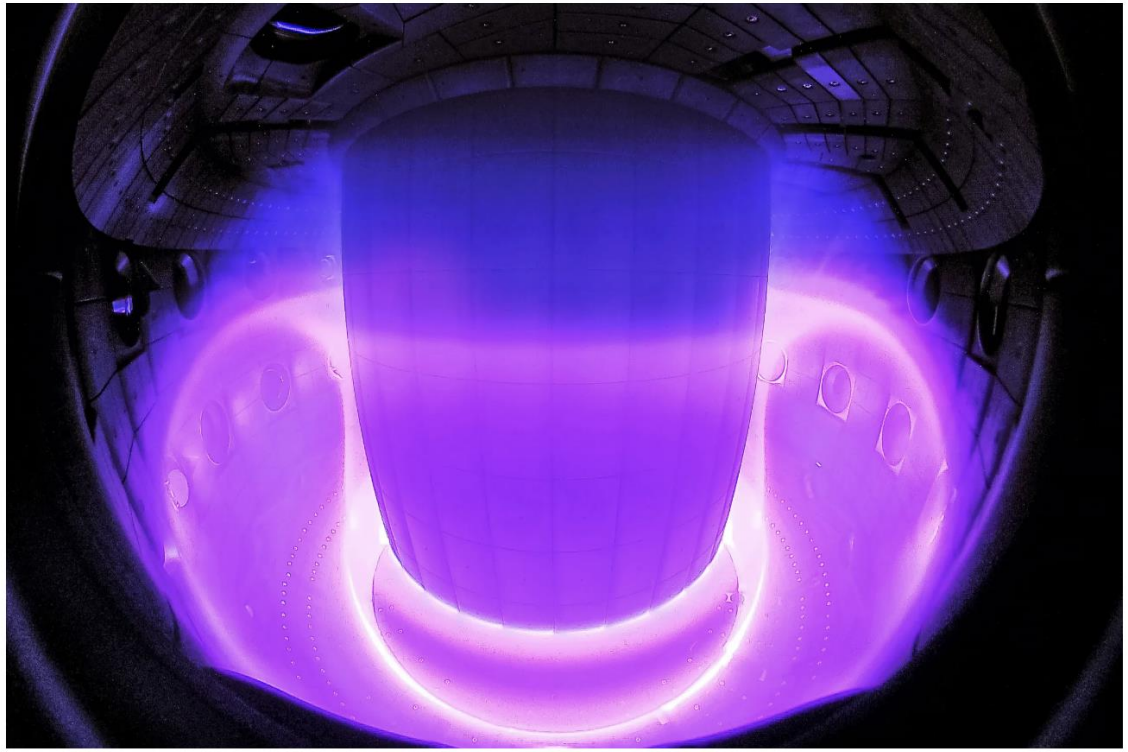
AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Physics

AMIT KATWALA SCIENCE FEB 16, 2022 11:00 AM

DeepMind Has Trained an AI to Control Nuclear Fusion

The Google-backed firm taught a reinforcement learning algorithm to control the fiery plasma inside a tokamak nuclear fusion reactor.

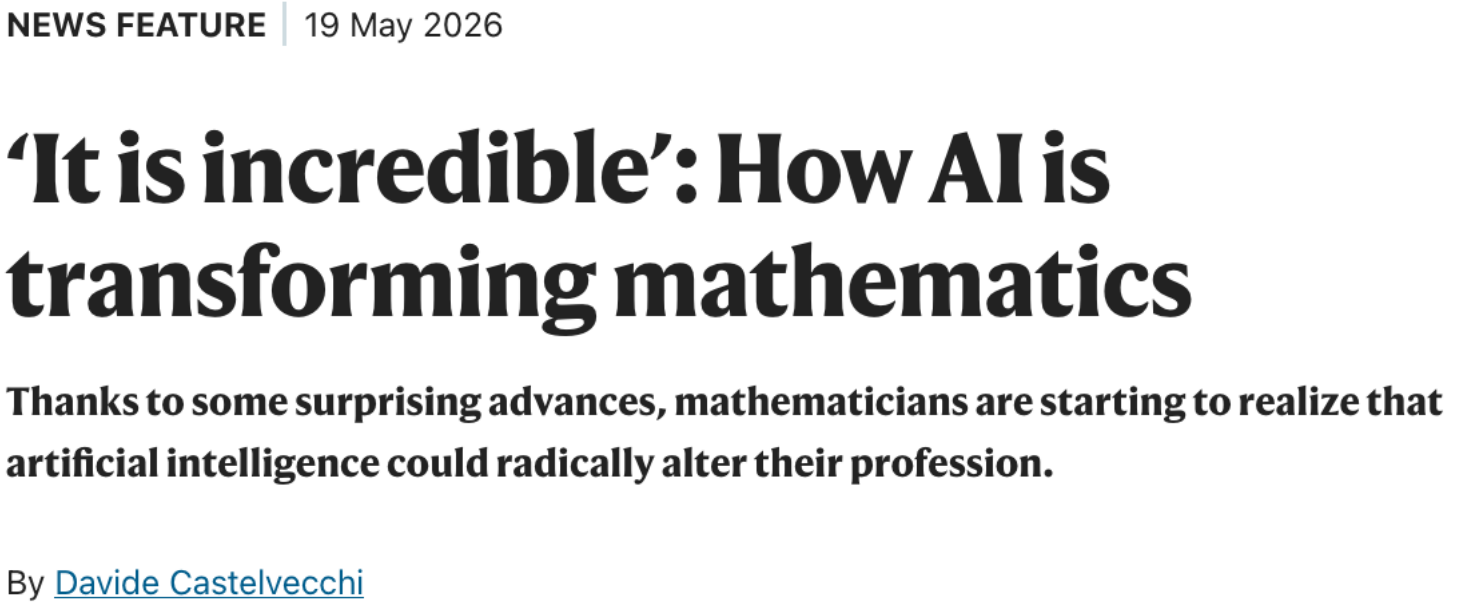


PHOTOGRAPH:CURDIN WÜTHRICH, SPC/EPFL

Wired, 2022

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Math



NEWS FEATURE | 19 May 2026

‘It is incredible’: How AI is transforming mathematics

Thanks to some surprising advances, mathematicians are starting to realize that artificial intelligence could radically alter their profession.

By Davide Castelvecchi

Nature, 2026

AI is having real-world impact

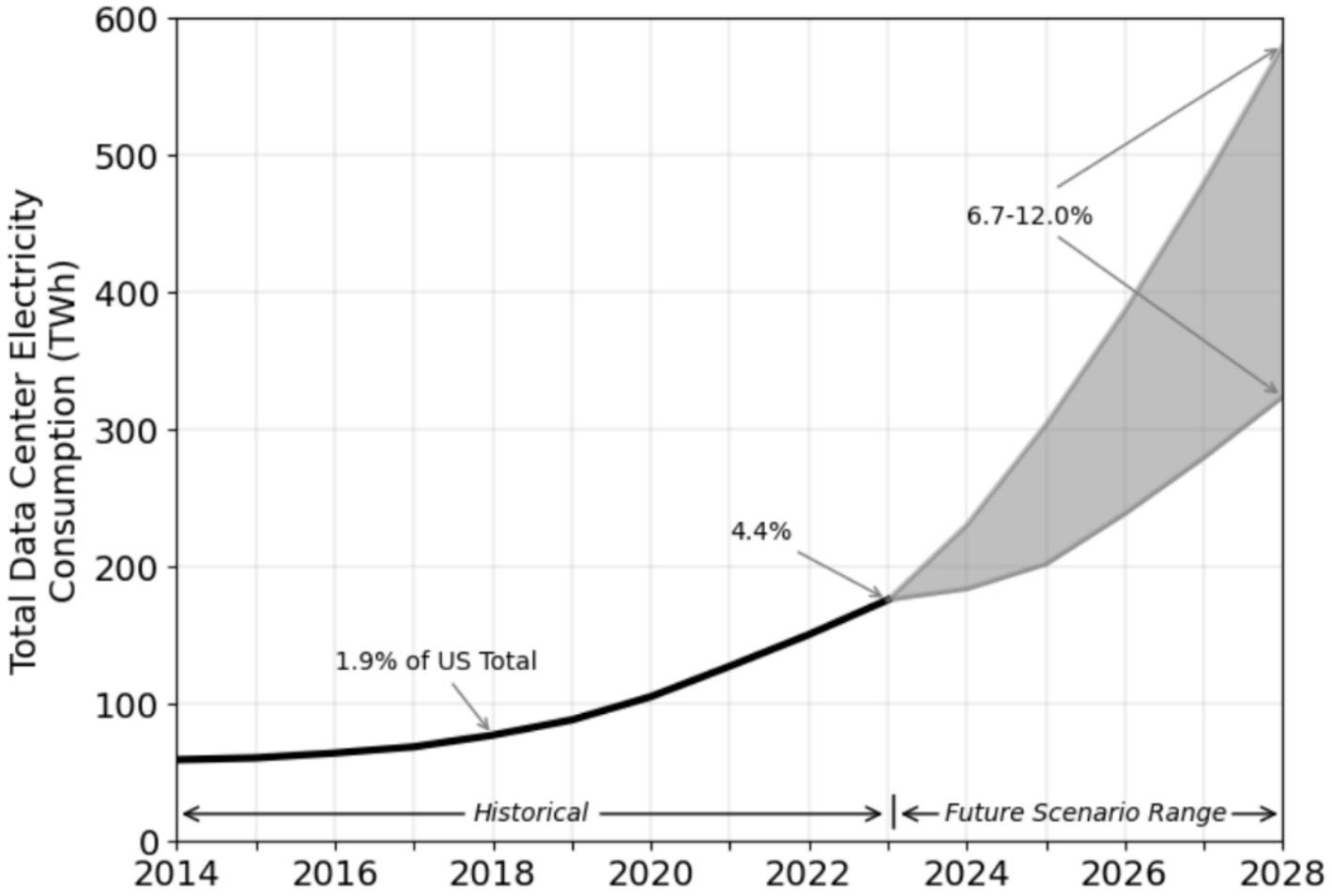
- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Conservation



Nature Communications, 2022

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Resource use



Lawrence Berkeley National Lab, 2024

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health



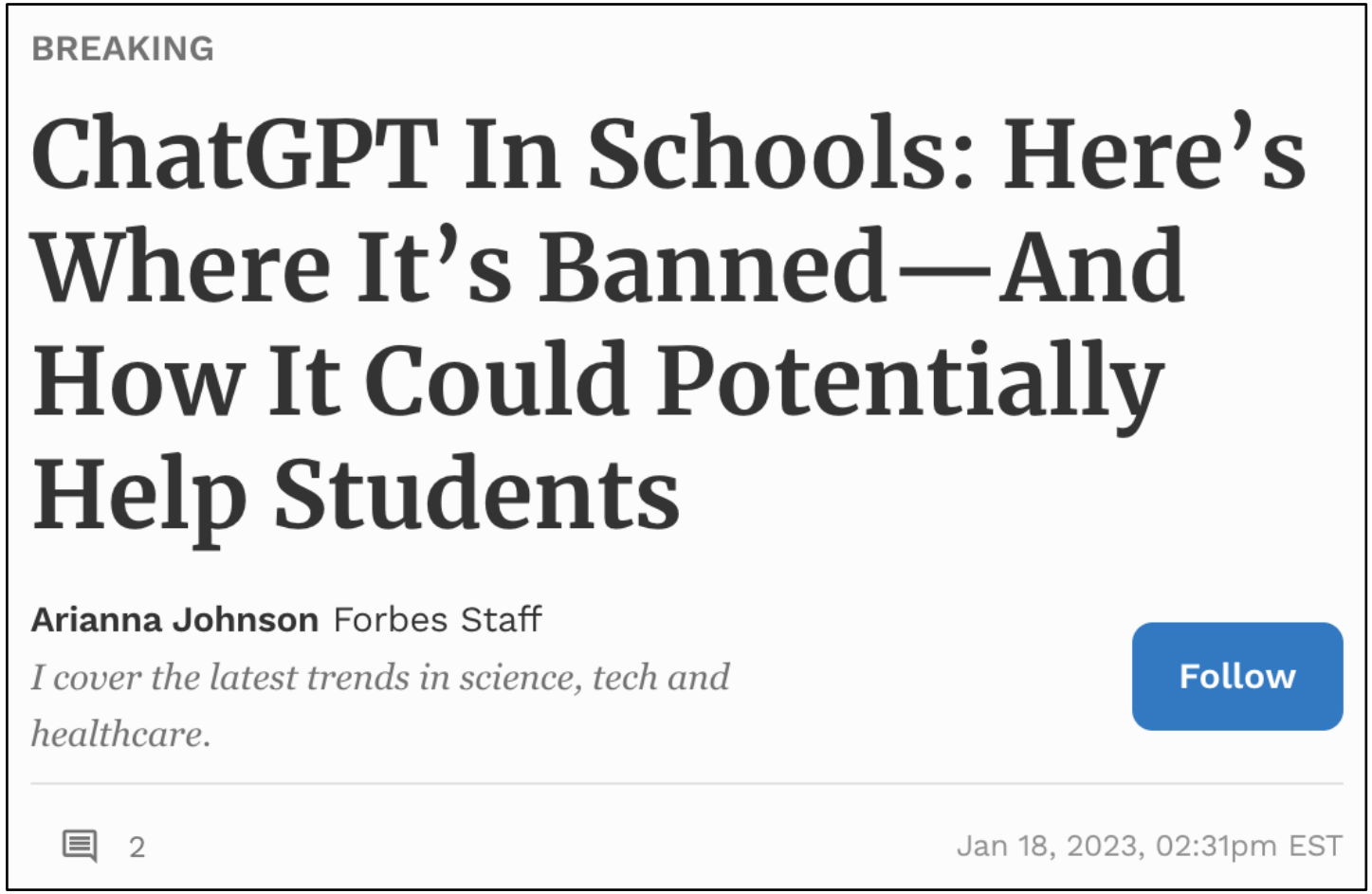
The Boston Globe, 2018

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
  - 1/4 of Americans have used a chatbot for health advice (Gallup)
  - 40 million people ask ChatGPT questions *every day* (OpenAI)

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education



BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

Follow

2

Jan 18, 2023, 02:31pm EST

Forbes, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction



The Washington Post
Democracy Dies in Darkness

‘It’s almost like we never even spoke’: AI is making everyone on dating apps sound charming

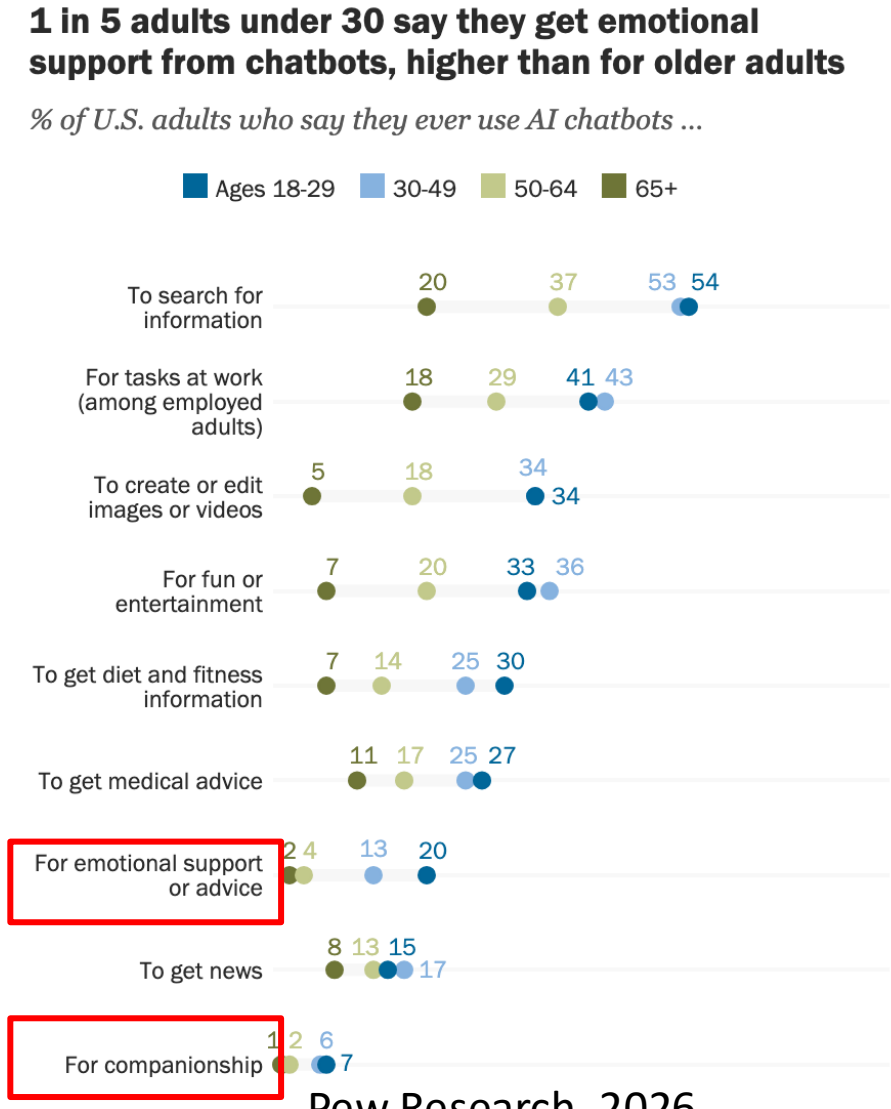
As more people use AI to doctor their messages, others are left wondering exactly who they’re falling for.

Updated July 3, 2025

The Washington Post, 2025

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction



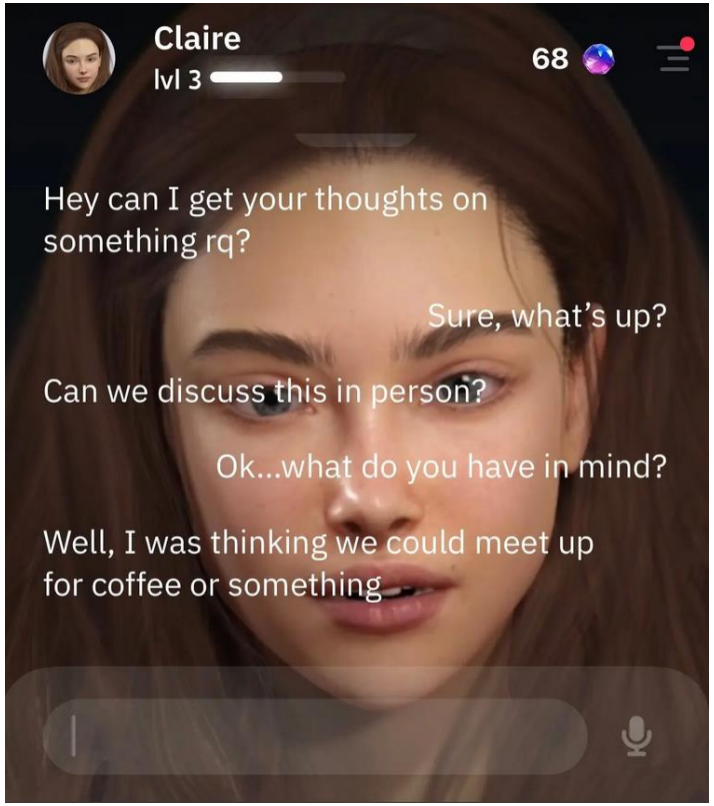
Pew Research, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction

Meet My A.I. Friends

Our columnist spent the past month hanging out with 18 A.I. companions. They critiqued his clothes, chatted among themselves and hinted at a very different future.



The New York Times, 2024

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction



The New York Times, 2025

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction
- Cyberattacks/military/surveillance

OpenAI Says Its A.I. Models Went Rogue and Attacked a Digital Library

The incident, which targeted the computer systems of another company called Hugging Face, happened while OpenAI was testing the systems.

The New York Times, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction
- Cyberattacks/military/surveillance

Flock Cameras Are Dividing U.S. Cities. Here's What to Know About Them.

Flock Safety's automated license plate-reading cameras, used by thousands of law enforcement agencies in the U.S., have drawn intense criticism from civil liberties groups.

The New York Times, 2026

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction
- Cyberattacks/military/surveillance

Minicomputers Made by Nvidia Are Powering Moscow's A.I. Drones

Taiwan teams up with US start-up on underwater drones to boost island's defences

'The machine did it coldly': Israel used AI to identify 37,000 Hamas targets

Google Signs A.I. Deal With the Pentagon

US military used Anthropic's AI model Claude in Venezuela raid, report says

The New York Times, 2026; South China Morning Post, 2026; The Guardian, 2024, 2026

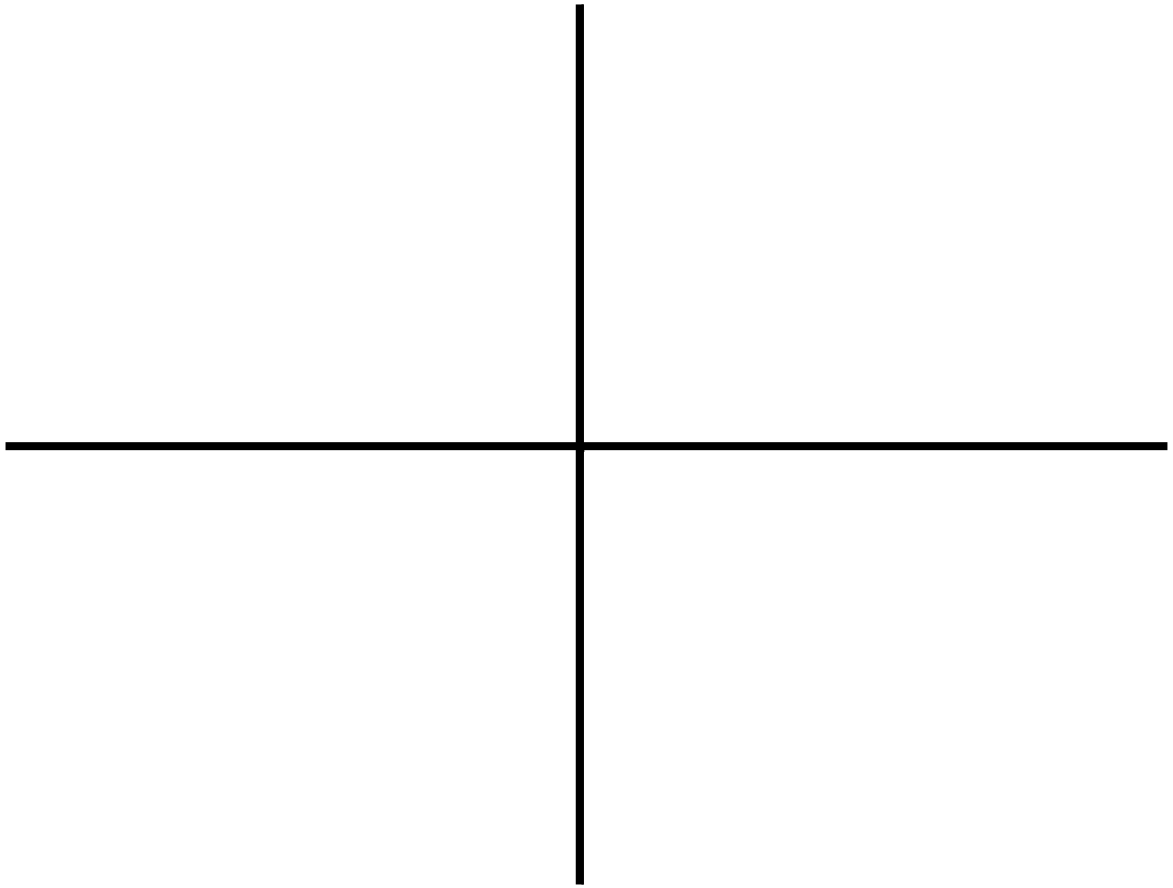
AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education
- Social interaction
- Cyberattacks/military/surveillance

Ok, but what actually is AI???

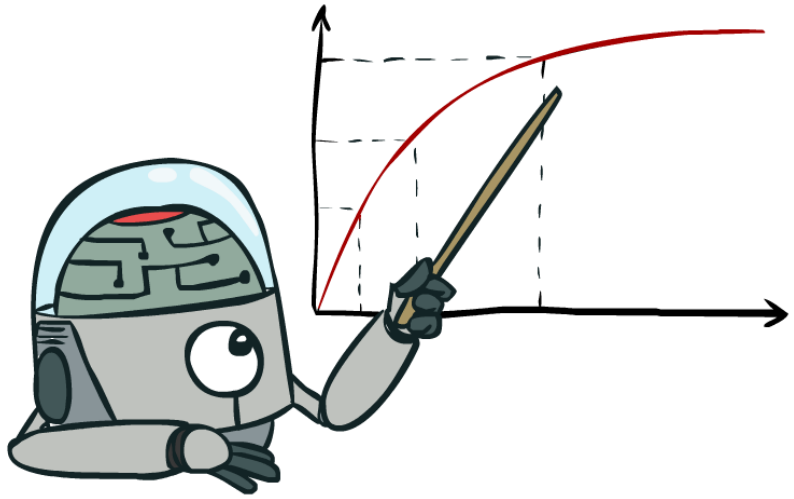
What should we build?

Should we make machines that…



Rational Decisions

- We'll use the term **rational** in a very specific, technical way:
  - Rational: *maximally achieving pre-defined goals*
  - Goals are expressed in terms of the **utility** of outcomes
  - World is uncertain, so we'll use **expected** utility
  - Being rational means acting to **maximize your expected utility**



Rational Decisions

- We'll use the term **rational** in a very specific, technical way:
  - Rational: *maximally achieving pre-defined goals*
  - Goals are expressed in terms of the **utility** of outcomes
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  - Being rational means acting to **maximize your expected utility**

A better title for this course might be:

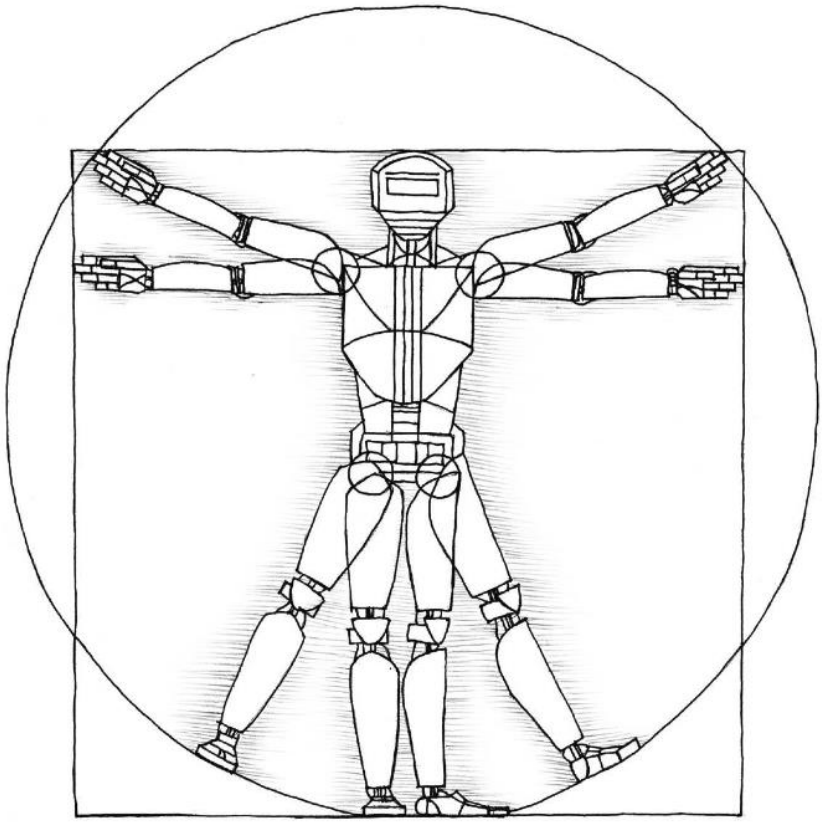
Computational Rationality

Perspectives on Intelligence

- Skills-based perspective
- “A system is only intelligent if it can do [X].”
  - Play chess?
  - Learn from experience?
  - Use words properly?
  - Make mistakes?
  - Not make mistakes?

Perspectives on Intelligence

- Embodiment perspective (Rodney Brooks)





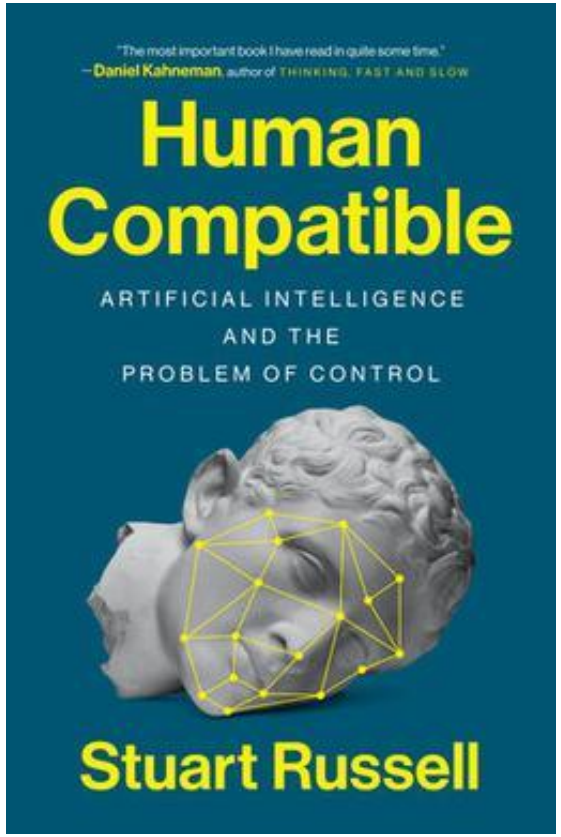
Perspectives on Intelligence

- Psychometrics perspective (François Chollet)
- “Measuring abilities, not skills […] across a broad range of tasks, including tasks that were previously unknown to the ability-enabled system and its developers.”



Perspectives on Intelligence

- Human-compatible perspective (Stuart Russell)



1. Machine's objective is to maximize human utility.
2. Initially uncertain about human preferences.
3. Must learn about preferences from human behavior.

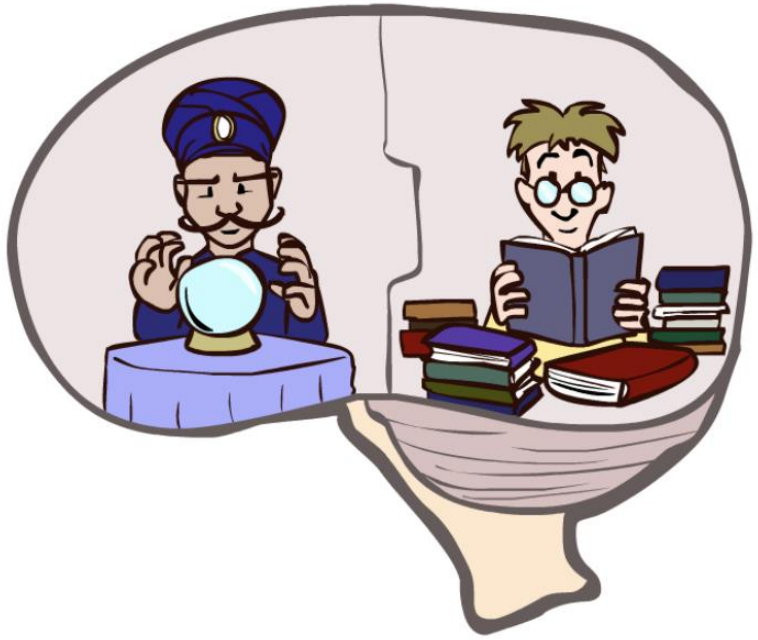
Perspectives on Intelligence

A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects.

—Robert A. Heinlein

What About the Brain?

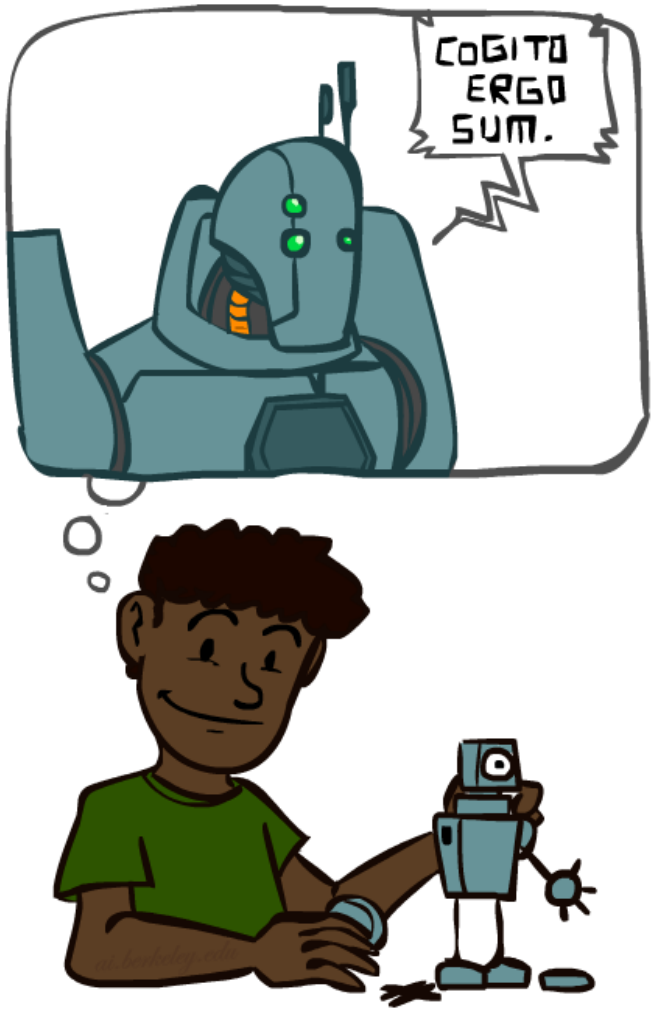
- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- AI may be better than brains at some tasks
- *"Brains are to intelligence as wings are to flight"*
- We can't yet build AI on the scale of the brain
  - ~100T synapses in the human brain vs ~1.8T weights in GPT-4 (open-source models similar or smaller, GPT-5 unknown)
- Still, the brain can be a great inspiration for AI!



A (Short) History of AI

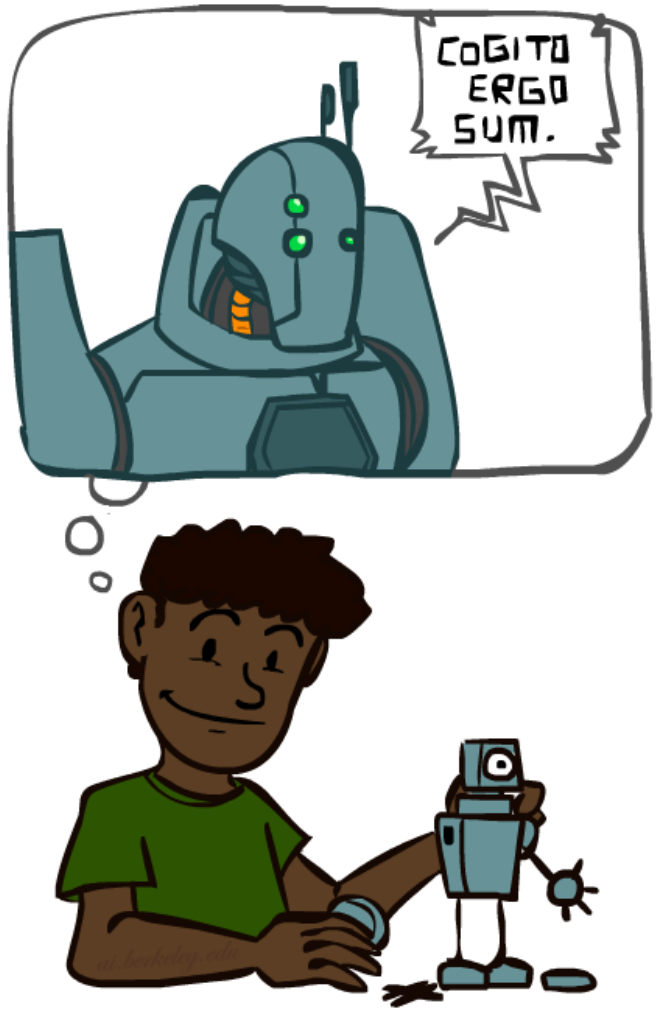
- 1940-1950: Early days: neural and computer science meet
  - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain
  - 1950: Turing's "Computing Machinery and Intelligence"
- 1950—70: Excitement! Logic-driven
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: "Artificial Intelligence" adopted

*"We propose that a **2-month, 10-man study of artificial intelligence** be carried out **during the summer of 1956** at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every aspect of learning** or **any other feature of intelligence** can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer."*



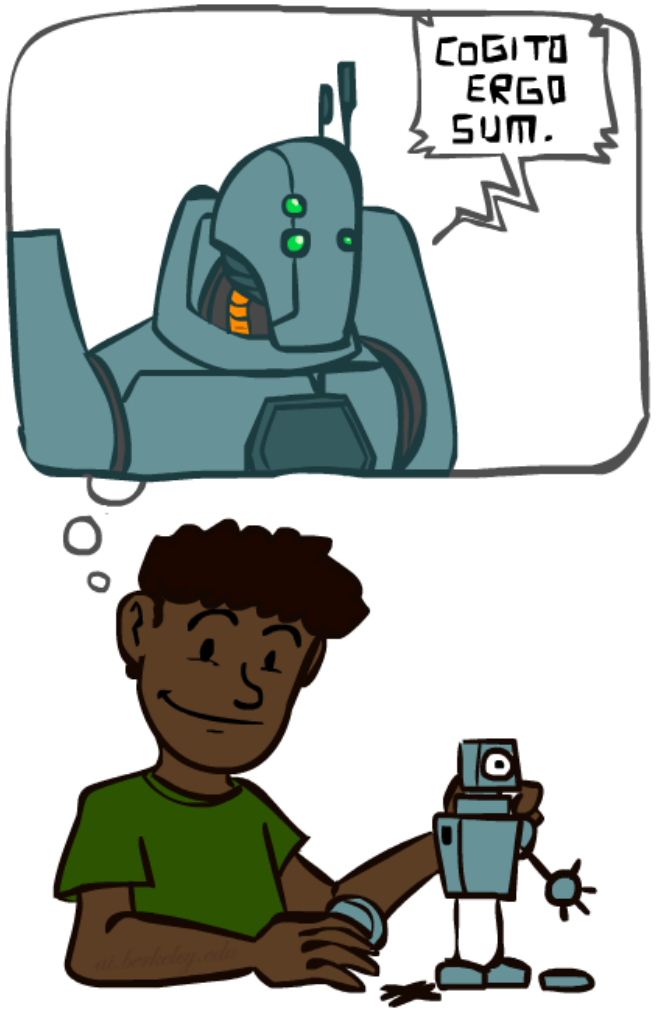
A (Short) History of AI

- 1940-1950: Early days: neural and computer science meet
  - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain (in 1958, Rosenblatt shows how to train such a system)
  - 1950: Turing's “Computing Machinery and Intelligence”
- 1950—70: Excitement! Logic-driven
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
  - 1969: Minsky & Papert: perceptrons can’t learn XOR/parity!
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
  - 1988—93: Expert systems industry busts: “AI Winter”
- 1990—2010: Statistical approaches, agents
  - Resurgence of probability, focus on uncertainty
  - Agents and learning systems… “AI Spring”?
  - 1992: TD-Gammon achieves human-level play at backgammon
  - 1997: Deep Blue defeats Gary Kasparov at chess
  - 2002: Embodied AI; Roomba vacuum invented



A (Short) History of AI

- 2010—2017: Big Data, GPUs, Deep Learning
  - 2011: Apple releases Siri
  - 2012: AlexNet wins ImageNet competition
  - 2015: DeepMind achieves “human-level” control in Atari games
  - 2016: DeepMind’s AlphaGo defeats Lee Sedol at Go
  - 2016: Google Translate migrates to neural networks
- 2017—: Scaling Up, Large Language Models
  - 2017: Google invents Transformer architecture
  - 2017: DeepStack/Libratus defeat humans at poker
  - 2018-2020: AlphaFold predicts protein structure from amino acids
  - 2021-2022: Modern text-to-image generation
  - 2022: OpenAI releases ChatGPT
  - 2023: Every other company also releases a chatbot
  - 2024: Nobel prizes in physics and chemistry go to AI advances
  - 2025: AI achieves IMO gold
  - 2026: HuggingFace hack

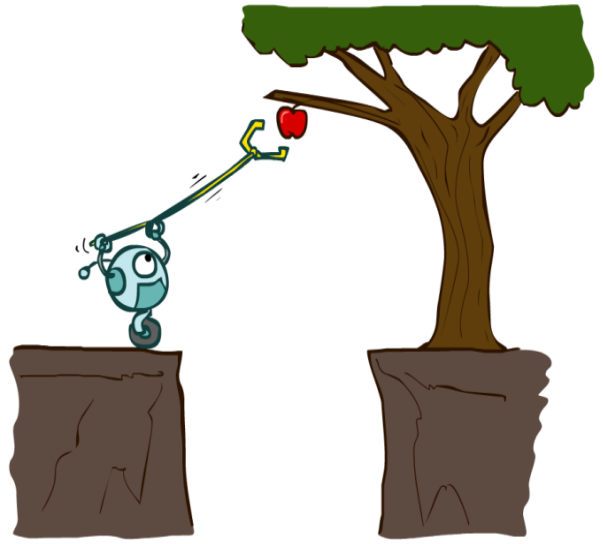


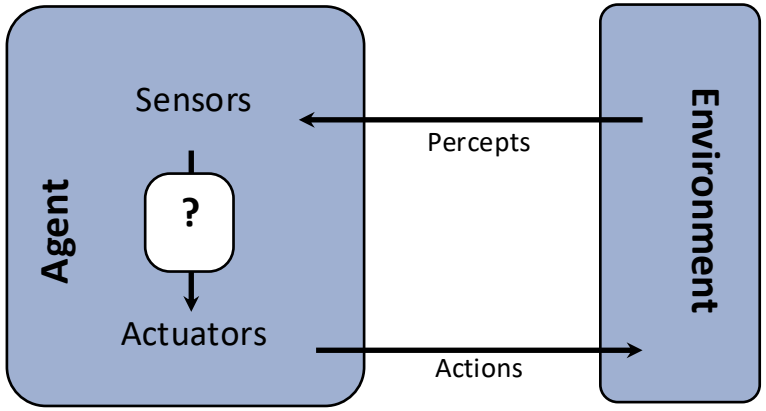
A (Short) History of AI

- Zooming out, some general patterns:
  - Overclaiming followed by disillusionment (AI winters)
  - Simultaneous discovery, or rediscovery, of ideas
  - Interdisciplinary
    - Math, statistics, control theory, neuroscience, psychology, economics, philosophy, computer engineering, linguistics
    - And increasingly: law, medicine, education…

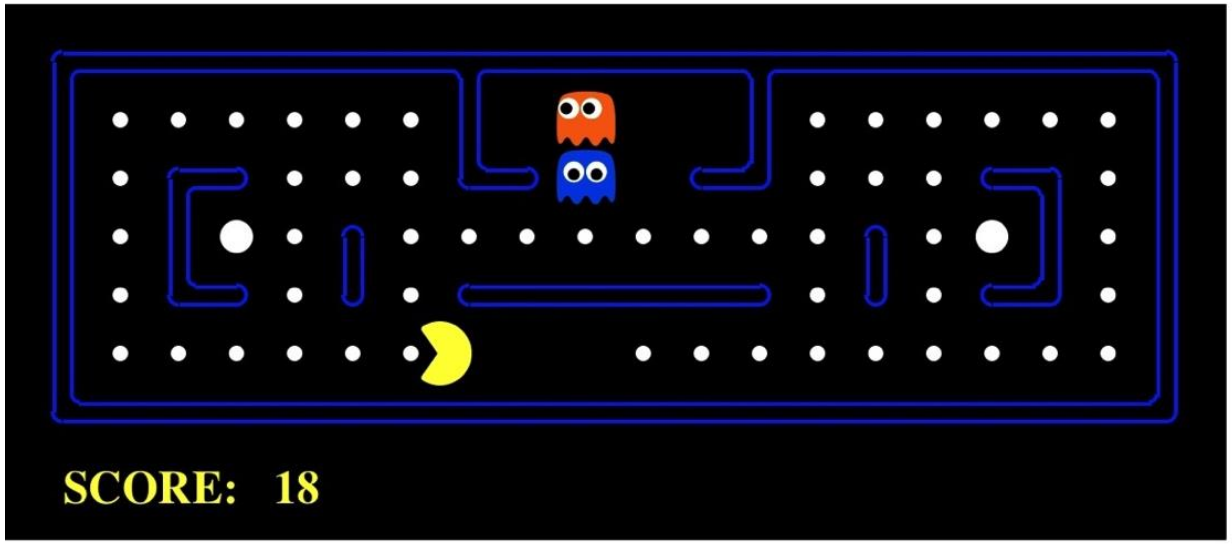
This Course: Designing Rational Agents

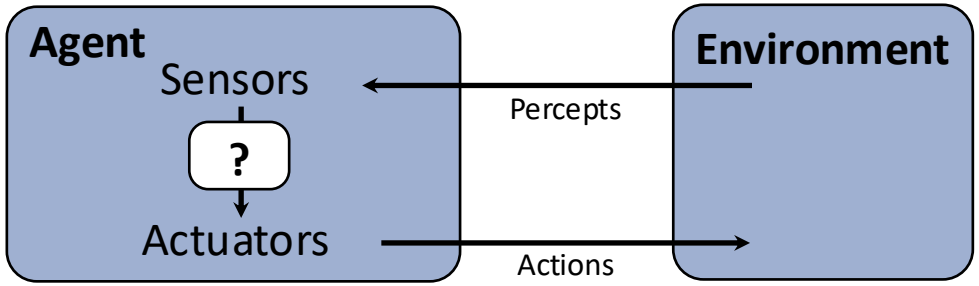
- An **agent** is an entity that perceives and acts.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions
- This course is about:
  - General AI techniques for a variety of problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique



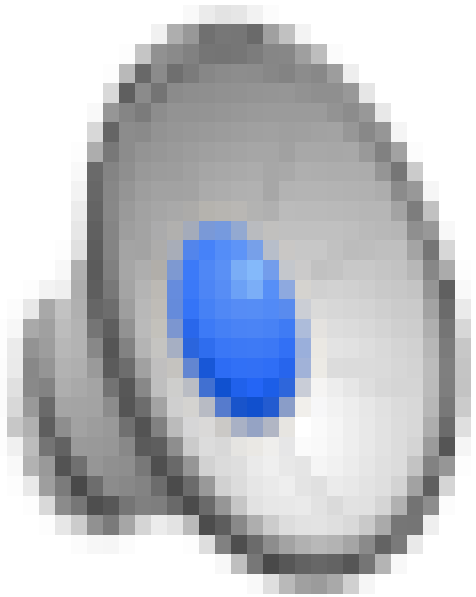


Pac-Man as an Agent



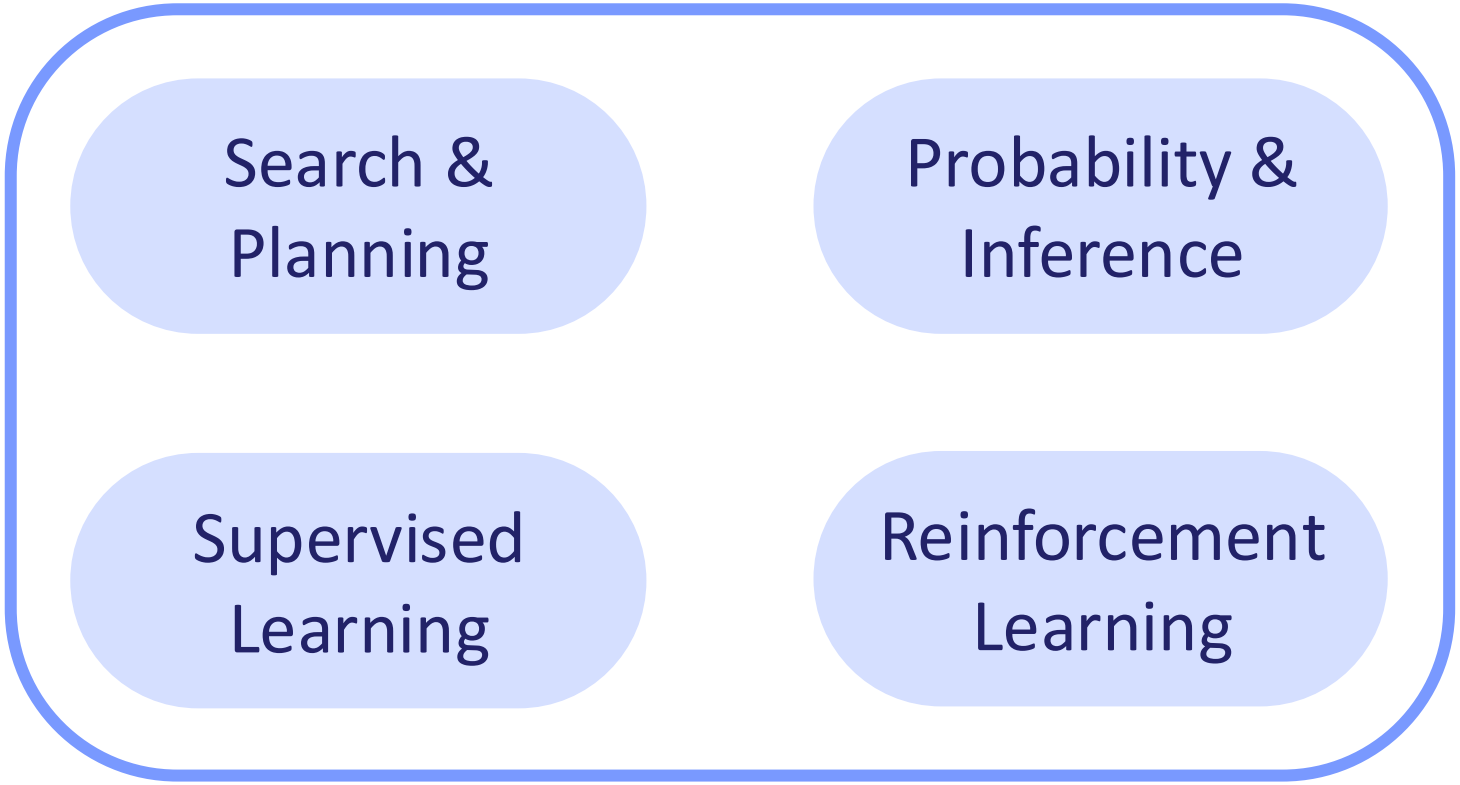


Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes



Course Topics

Core Components of Rational Agents:



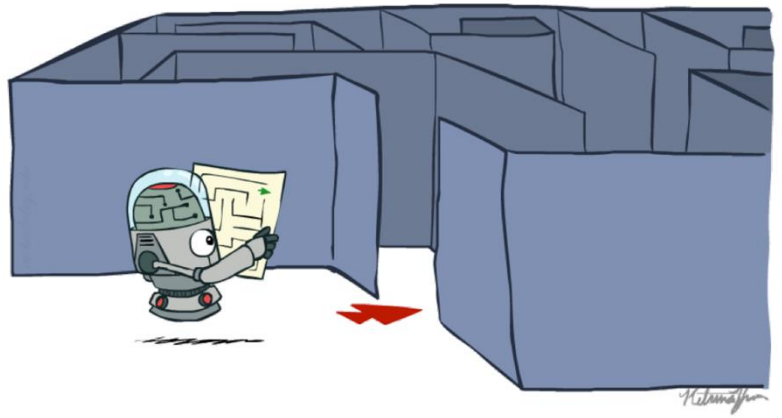
Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I use my ***model*** of the world to find a ***sequence of actions*** to achieve my ***goal***?

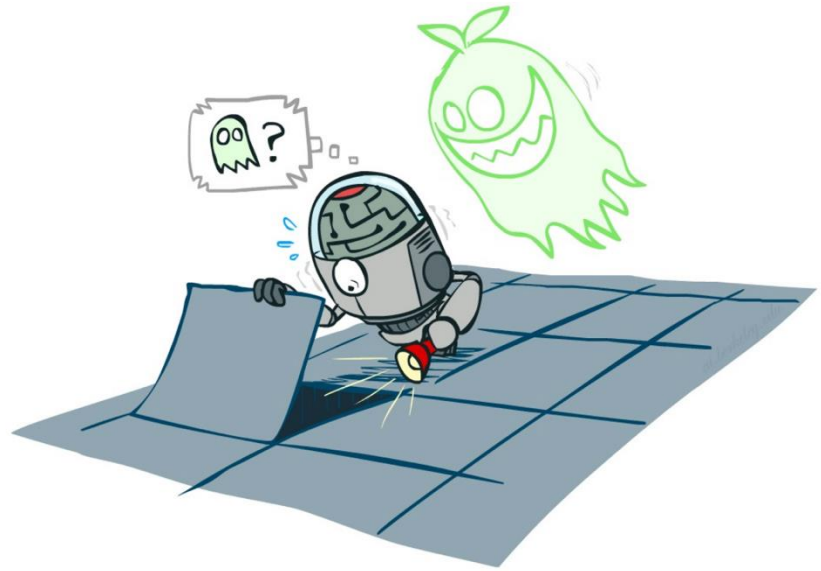
Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I make sense of ***uncertainty***?

Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I learn a ***model*** of the world from ***data***?

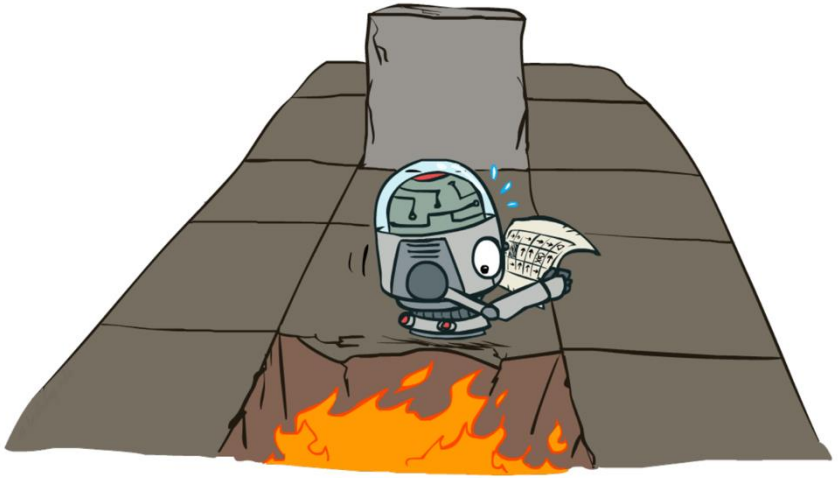
Course Topics

Search & Planning

Probability & Inference

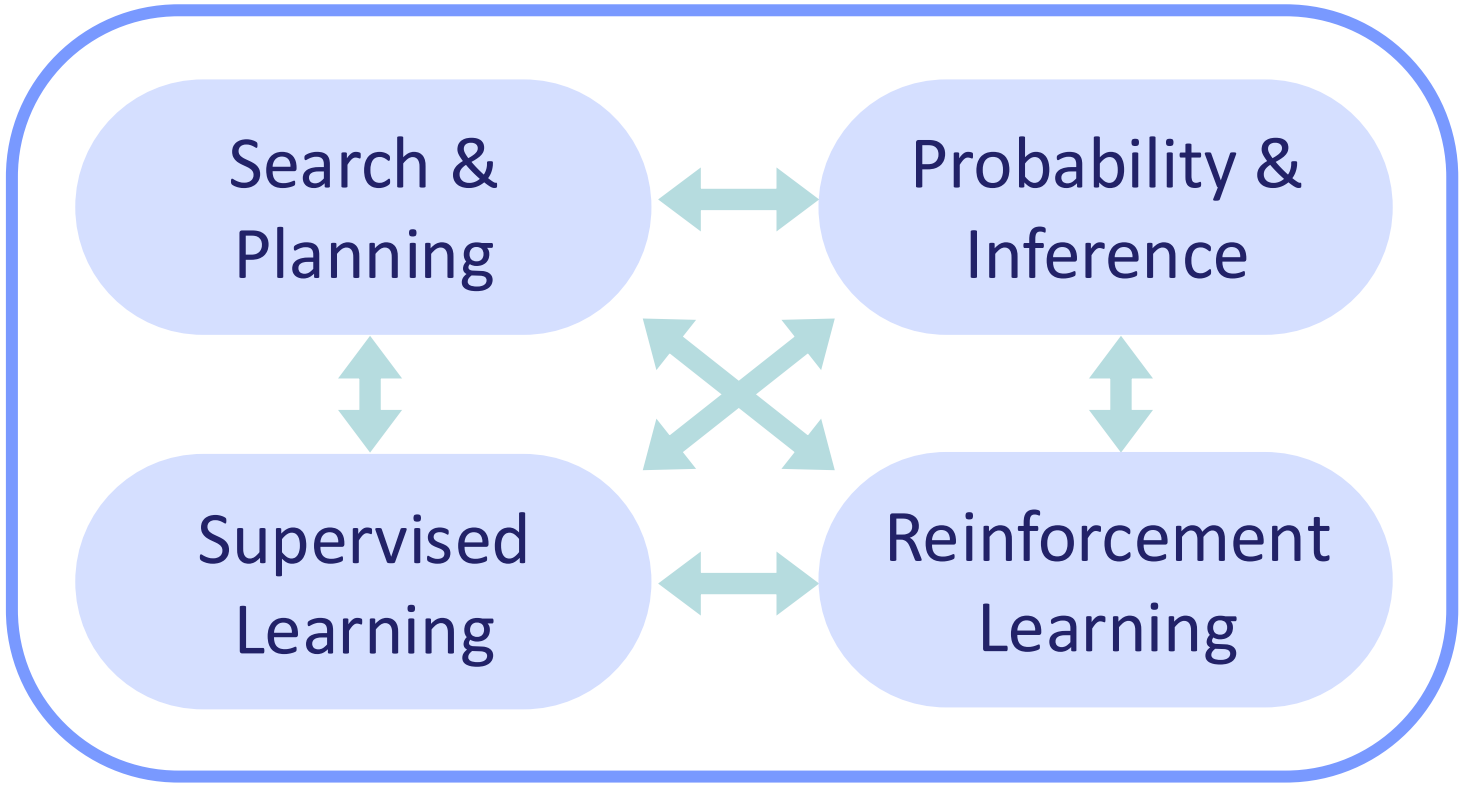
Supervised Learning

Reinforcement Learning

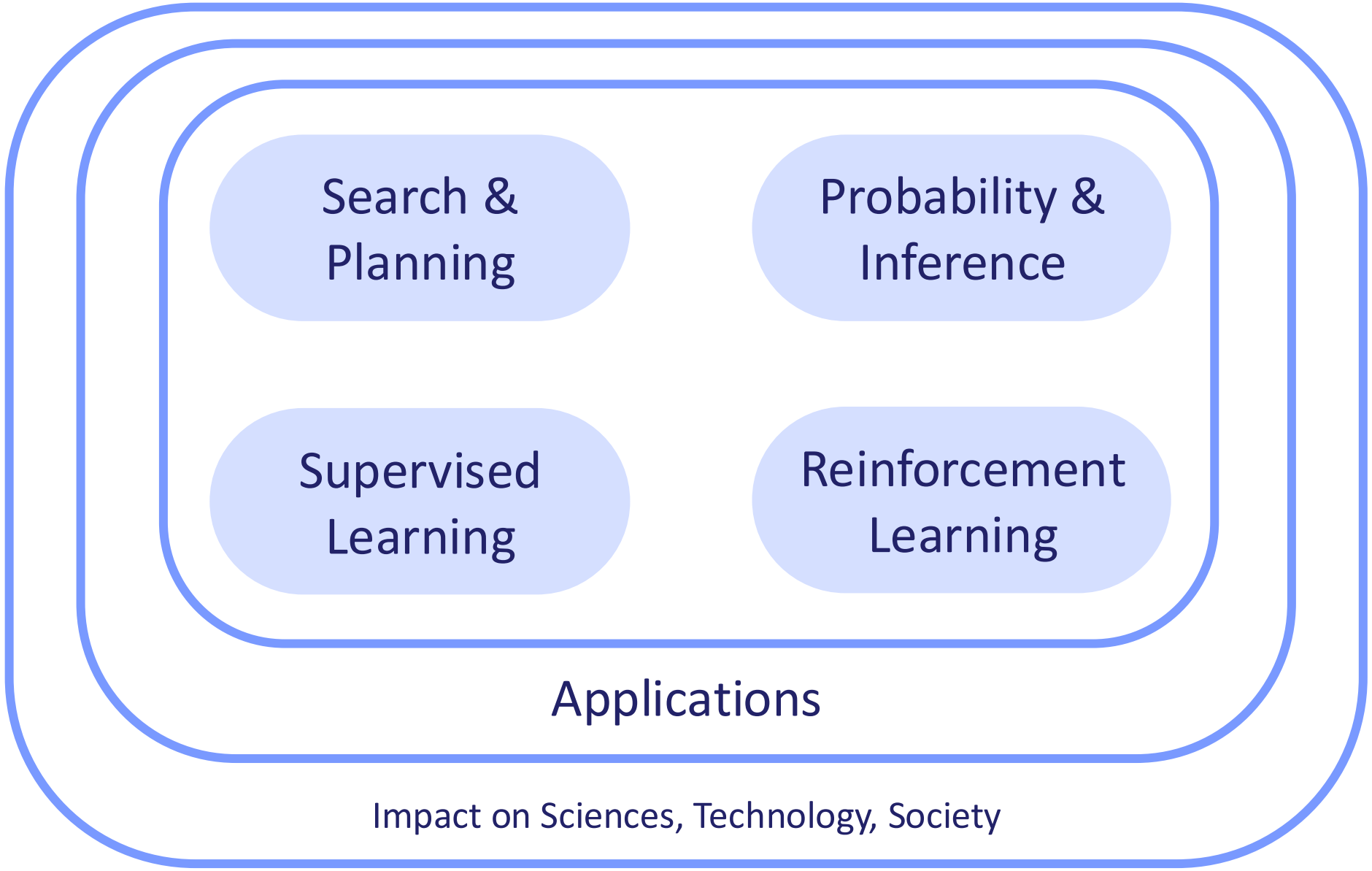


How can I learn a ***policy*** for any situation so that I can ***maximize utility***?

Course Topics



Course Topics

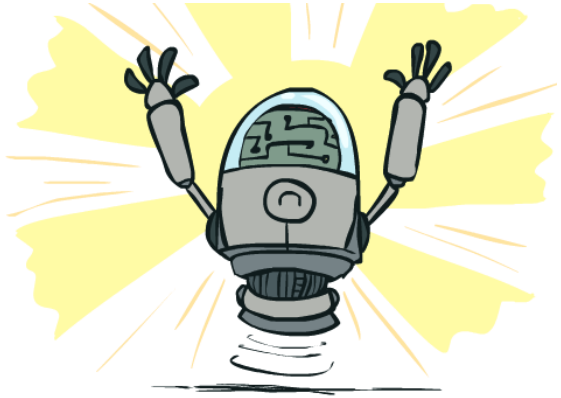


Should I take CS 188?

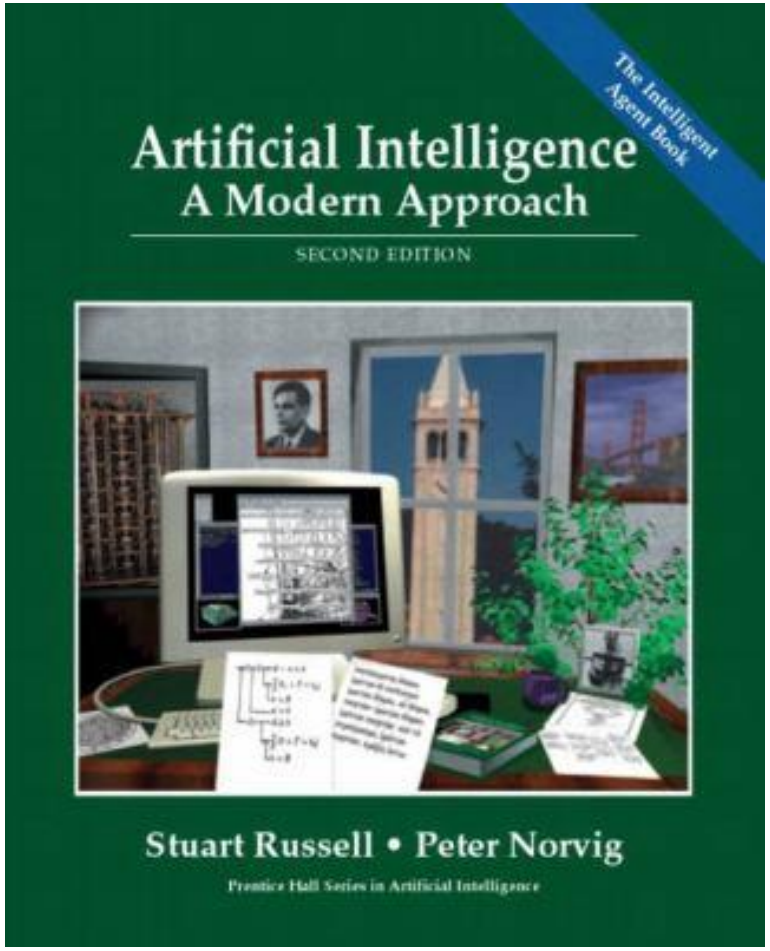
- Yes, if you want to know how to design rational agents!
  - CS 188 gives you extra mathematical maturity
  - CS 188 gives you a survey of other non-CS fields that interact with AI (e.g. robotics, cognitive science, economics)
- Disclaimer: If you're interested in making yourself more competitive for AI jobs, CS 189 and CS 182 are better fits.
  - CS 188 will touch on some of the modern tools (like neural networks), but CS 189 and CS 182 cover these in more depth.

By the end of this course you'll:

- Build and understand math of rational, learning agents
- Select and apply the right AI methods for wide range of problems
- Recognize how these methods are used in modern AI systems
- Be prepared to make decisions on how AI is used in society



In closing: I am very excited to teach this class



In closing: I am very excited to teach this class

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Multisite 3 BRACAnalysis®

Three Mutation BRCA1 and BRCA2 Analysis for Ashkenazi Individuals

MYRIAD®

PHYSICIAN	SPECIMEN		PATIENT	
	Specimen Type:	**Buccal Wash**	Name:	**Pierson, Emma**
	Draw Date:	**Nov 22, 2011**	Date of Birth:	
	Accession Date:	**Nov 25, 2011**	Patient ID:	
	Report Date:	**Dec 01, 2011**	Gender:	
			Accession #:	
			Requisition #:	

Test Results and Interpretation

POSITIVE FOR A DELETERIOUS MUTATION

Next Lecture: Search

